

**WEB-BASED 3D PRINTING SERVICE ORDERING
MANAGEMENT SYSTEM FOR TECHWORKZ**

**A Capstone Project
Presented to the Faculty of
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**In Partial Fulfillment
Of the Requirements for the Degree of
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**

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ABSTRACT

This examines the development of a Web-based 3d Printing Service Ordering Management System for TechWorkz, aimed at making the process simpler and more efficient from customer order entry through printing. The system seeks to streamline customer order processing, make it easier to communicate with clients, and improve with these goals in mind, it solves the existing problems such as slow response speed, varied prices, and no records. A quantitative approach was used in collecting user feedback and obtaining information about system performance. A 5-Point Likert Scale was used as the core tool in assessing how usable and functional the system is. Overall, results showed that both the opinions of clients and staff were broadly consistent. It was broadly agreed that the system met quality standards and should achieve all recommendations made about functionality, reliability, and effectiveness. The proposed system is based on cloud technology. This will enable clients to place an order, trace its status, and communicate with the company's staff. Automation is incorporated, with a view to reducing workload while improving quality control. Key findings suggest that the web-based system can significantly enhance customer satisfaction by providing both a centralized platform for order management and an area in which people can talk to each other. The cloud-based design is scalable and with automation added into manual work, efficiencies will be greater as well as customer experiences enhanced. All of these directions point to an operational system designed. In summary, the Web-based 3D printing service ordering management system for TechWorkz promises to redefine the company's operations. The system provides an integrated approach to order management and communication, and is thus a model for other Web-based 3D printing service companies looking to modernize their operations.

Keywords: cloud-based technology, additive manufacturing technologies, supply chain management, e-commerce.

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CHAPTER I

PROJECT AND ITS BACKGROUND

3D printing, also known as additive manufacturing (AM), has emerged as a revolutionary technology in manufacturing and design. By adding material layer by layer based on digital designs, 3D printing allows for reduced waste, enhanced customization, and the production of complex geometries that would be challenging with traditional subtractive manufacturing methods.

TechWorkz 3D Printing Services is a business leveraging this technology to provide unique solutions to its clients. To improve the customer experience and operational efficiency, TechWorkz aims to implement a Web-based 3D Printing Service Ordering Management System. This system is intended to streamline the ordering process, improve customer communication, and enhance the management of sales and payment records.

The proposed system addresses key issues such as slow response times to customer inquiries, inconsistent pricing, and disorganized record-keeping. By centralizing these processes in a web-based platform, TechWorkz can reduce response times, better manage sales and payment records, and improve customer satisfaction through a more seamless ordering process. Additionally, improved communication through the system's chat functionality and real-time updates can foster a stronger relationship between TechWorkz and its client.

Purpose and Description

TechWorkz 3D Printing Services is a highly regarded company specializing in advanced 3D printing technologies. Located in Sunnycrest Village, Salitran II, Dasmarinas City, Cavite, TechWorkz was established in 2019 by Mr. Alfred Magno. Since its inception, the company has built a strong reputation for its commitment to innovation, precision, and high-quality service. The company operates with five state-of-the-art 3D printers and is supported by a skilled team of five dedicated professionals, including a 3D artist knowledgeable in 3D modeling and customization. TechWorkz offers a diverse array of services, including rapid prototyping, custom manufacturing, and the production of unique collectibles, souvenirs, and functional parts tailored for various industries.

The Company's success can largely be attributed to its unwavering focus on customer satisfaction and its dedication to delivering exceptional quality. The company's approach to 3D printing is inherently customer-centric, ensuring that each project is meticulously customized to meet the specific needs and requirements of the client. Despite its significant achievements and growth, TechWorkz encounters several notable challenges within its current operational framework. One of the primary issues is the reliance on Facebook Messenger for handling customer inquiries. This communication method has led to considerable inefficiencies and delays, making it increasingly difficult to track and manage customer requests effectively. The absence of a centralized system for record-keeping adds to these challenges, resulting in the frequent misplacement of crucial information and extended response times to customer inquiries.

Another critical challenge faced by TechWorkz is the lack of a standardized pricing system. The current method for cost estimation of 3D printing orders is prone to errors, which can lead to significant issues such as overcharging or undercharging clients. This inconsistency in pricing not only affects customer satisfaction but also poses a potential threat to the company's overall profitability. Additionally, the existing process for monitoring and tracking the progress of 3D printing projects is inadequate. Clients currently need to contact the staff for updates, which increases the workload for the team and can result in delays in meeting project deadlines.

In response to these challenges, the proponent proposed the implementation of a Web-based 3D Printing Service Ordering Management System. This new system is designed to significantly enhance efficiency by streamlining customer inquiries, order processing, and internal operations. With this system in place, clients will have the ability to place orders, track the progress of their 3D prints, and make payments through a centralized online platform. The primary objective of implementing this system is to improve overall customer satisfaction, boost operational efficiency, and substantially reduce the risk of errors in order processing and record-keeping.

The proposed system will also include a centralized repository for managing customer inquiries, which will enable the staff to handle and respond to requests more effectively and promptly. An automated cost estimation feature will address the issue of pricing errors, providing clients with accurate and consistent cost information. Additionally, the tracking functionality will allow both clients and staff to monitor the status of orders in real-time, thereby improving communication and reducing the need for follow-up inquiry.

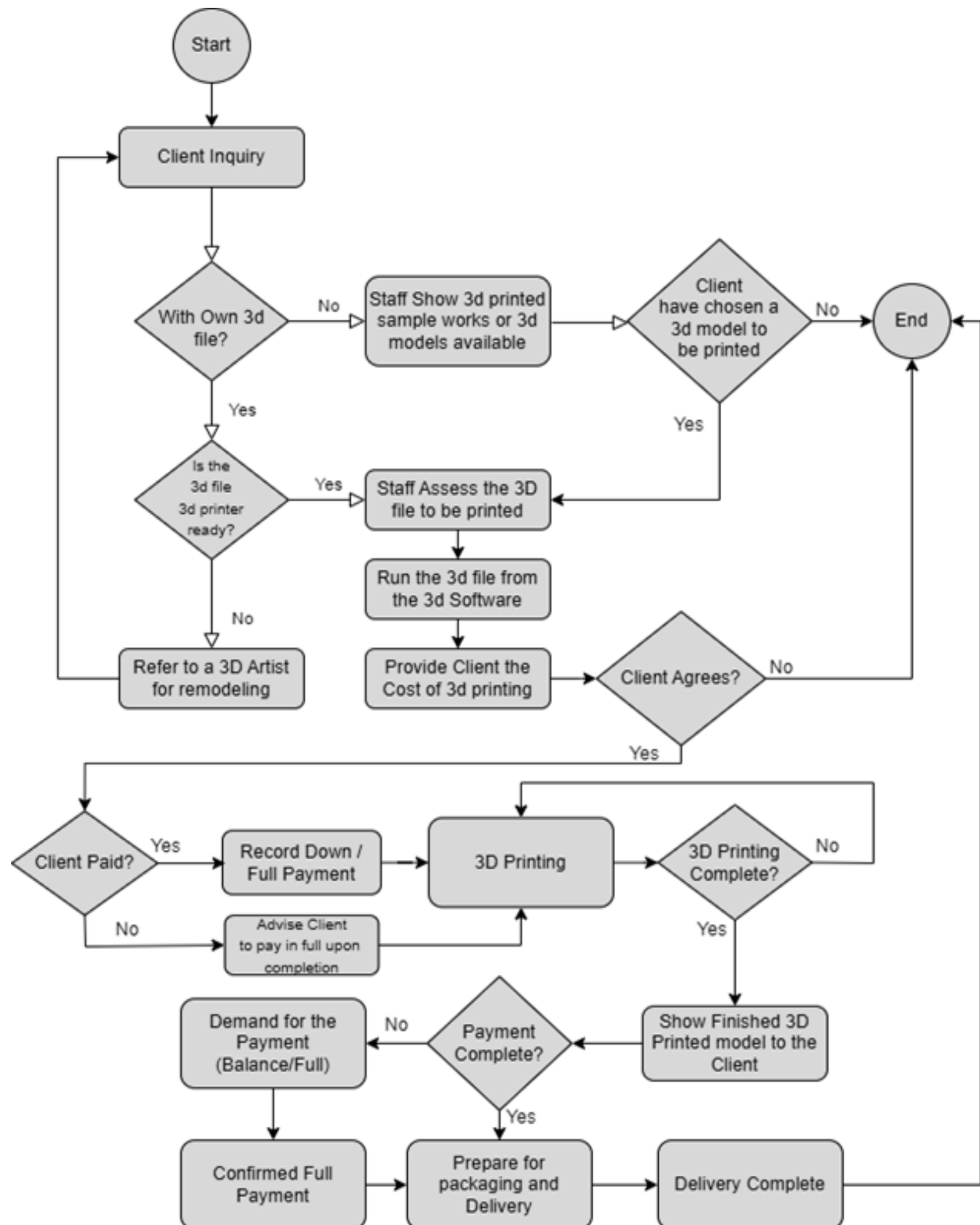
Problems in the Existing Business Process

TechWorkz's reliance on Facebook Messenger for client inquiries has introduced significant inefficiencies into its business process. Since staff must manually track customer inquiries, this system creates longer response times and disorganized record-keeping. The lack of a centralized platform for managing inquiries often leads to the misplacement of crucial information, resulting in delays and missed opportunities to engage with clients effectively. This manual approach also complicates efforts to maintain records of previous works, making it challenging for staff to present clients with sample models or highlight previous projects, impacting customer satisfaction.

Another critical issue in TechWorkz's current process is the absence of a standardized pricing system for 3D printing services. Estimating the cost of a 3D print is often subjective and prone to error, leading to either overpricing or underpricing. Overpricing may deter potential clients, while underpricing can adversely affect profitability. This inconsistency creates a need for a reliable, automated system to standardize pricing and ensure accuracy, providing both customers and staff with clear expectations. These issues point to the need for a centralized, web-based system to improve efficiency and communication. The proposed Web-based 3D Printing Service Ordering Management System aims to solve these problems by providing a platform where clients can inquire, place orders, and track the progress of their 3D prints.

Figure 1

Existing Business Process Flow Diagram



Existing Company Business Process Flow Diagram as shown in Figure 1 , the process starts when a potential client reaches out through Facebook Messenger, inquiring about a 3D printing service. This can include inquiries about custom designs, materials, or pricing. After understanding the client's requirements, the staff may share sample 3D files or showcase previous projects to give the client an idea of the work quality and design capabilities. If the client has a 3D model file to be printed, it is shown to the staff for assessment. Once the client understands the 3D printing capabilities, the staff reviews the specific requirements, runs the 3D file through the printing software, estimates the cost based on material, size, printing time, and complexity, and provides a quotation.

Upon receiving the quotation, the client reviews the cost and terms. If they agree, they confirm the order and make the payment. Payment methods may include online transactions, such as cashless apps like Gcash or Maya. Whether a down payment, full payment ahead, or payment after completion is agreed upon, the staff proceeds with the 3D printing process. This involves loading the design file into the 3D printer and overseeing the printing to ensure quality and accuracy.

After printing, the staff performs a quality check to ensure the finished product meets the client's specifications and standards. Photographs of the finished item are taken as confirmation to the client that the product is ready for delivery. The staff then verifies payment status. If there is a balance, payment is requested from the client. The staff arranges for delivery, which could involve shipping through a courier service or arranging a pickup, with common options including Lalamove, Grab, LBC, or any preferred courier. The delivery and client confirmation mark the end of the transaction. The client receives the 3D-printed item, and the business ensures customer satisfaction.

Project Context

To address the issues outlined in the existing business process, the proposed Web-based 3D Printing Service Ordering Management System aims to provide TechWorkz with a streamlined, centralized platform for managing customer inquiries, orders, and internal processes. This system will serve as a comprehensive solution to the problems related to inefficient communication, inconsistent pricing, unorganized record-keeping, and lack of proper monitoring for 3D printing progress. The implementation of this system will create a more cohesive workflow for the company and improve the overall customer experience.

The proposed system consists of several modules, each designed to address specific issues identified in the current process. Below are the key modules and their functionalities:

1. **User Registration and Login Module:** This module allows clients to create accounts, log in, and access their order history. It ensures secure access to the system while providing a platform for customers to manage their information. An admin panel is included to manage user accounts, allowing the system administrator to create accounts for the owner and staff.
2. **Guest Account Module:** This module is for clients who want to browse the system without registering. It provides limited access, allowing guests to view the gallery, read FAQs, and explore the ready-to-print 3D models. However, placing an order requires account registration, ensuring a secure and organized process.
3. **Dashboard Module:** This module serves as the central hub for users. It includes a photo gallery of completed projects, a catalog of ready-to-print 3D models, and access to the ordering system. It aims to shift client inquiries from Facebook Messenger to a more structured platform, allowing for more efficient customer interaction and providing comprehensive information about Company's services.

4. Chat Module: To improve communication between clients and staff, this module allows for real-time chat within the system. This feature helps address customer queries efficiently and reduces the need for clients to use external communication platforms like Facebook Messenger.
5. Email Notification Module: This module sends automated email notifications to users during key events, such as registration, order placement, and payment confirmation. It enhances communication and keeps clients informed throughout the ordering process.
6. Photo Gallery Module: This module allows staff to upload photos of previous 3D printing projects, providing clients with visual examples of the company's work. It helps clients understand the quality and range of services offered by the Company.
7. Catalog of Ready-to-Print 3D Models Module: This module presents a list of pre-designed 3D models that clients can order directly, with corresponding prices and product details. It simplifies the ordering process for customers who prefer ready-made options.
8. Customized 3D Print Module: For clients who have specific requirements, this module allows them to customize 3D print orders by uploading their 3D model files in .stl or .obj formats. It also offers options for customizing size, color, and other attributes.
9. 3D Printing Calculator Module: To address pricing inconsistency, this module provides an automated cost estimation tool for customized 3D prints. It calculates costs based on model details, size, quantity, material type, and other factors, ensuring accurate and standardized pricing.
10. Payment Monitoring Module: This module enables staff to track client payments, whether full or partial. Clients can upload proof of payment via email or pay directly

through the system via Gcash or Paymaya using Paymongo as the gateway and staff will verify and update the status within the system and updates the status of an order. This module ensures a clear record of transactions and reduces confusion regarding payments.

11. 3D Printing Progress Status Module: To improve order tracking, this module allows clients to view the status of their 3D printing orders. It provides updates on key stages such as "Printing in Progress," "Estimated Time Left," and "Successfully Printed." This feature minimizes follow-up inquiries and keeps clients informed throughout the process.

12. Reports Generation Module: This module generates various reports, such as lists of client orders, sales reports, and completed orders. These reports help the owner and staff track business performance and make informed decisions.

Together, these modules aim to create a cohesive and efficient system that addresses the key problems TechWorkz has faced in its current business process. By implementing this system, the company can improve its customer communication, streamline order management, and ensure accurate record-keeping, leading to a more robust and customer-centric 3D printing service.

Objectives of the Study

The general objective of this study is to develop the "Web-based 3D Printing Service Ordering Management System for TechWorkz," which will enhance the efficiency and organization of the company's order management processes. The implementation of this system aims to improve customer interaction, streamline the order process, and provide a more effective solution to the problems encountered in the existing business model.

Specifically, the study aims to:

1. facilitate the transition from Facebook Messenger-based inquiries to a dedicated web-based platform, allowing clients to easily access information, place orders, and receive updates on their 3D printing projects.
2. provide a centralized platform where staff can manage customer inquiries, track orders, and oversee the progress of 3D printing projects.
3. process customer orders efficiently by automating the steps involved in 3D printing, from initial inquiry to final delivery, reducing the need for manual intervention and improving accuracy.
4. generate various reports, such as a list of client orders, sales reports, and completed orders, to help the business owner and staff make informed decisions about the company's operations and growth.
5. evaluate the system using the 5-point Likert Scale method to assess its functionality, usability, performance, and user feedback, ensuring that the system meets the needs of both clients and staff.

Significance of the Study

The significance of the study encompasses multiple stakeholders, including TechWorkz, its clients, staff, IT administrators, the broader 3D printing industry, and even potential guests or casual users of the system. This section discusses the importance of the proposed "Web-based 3D Printing Service Ordering Management System for TechWorkz" and how it benefits each group.

For the Specific Industry: The 3D printing industry stands to gain from the implementation of a robust web-based ordering system that improves efficiency and customer service. This system serves as a benchmark for other companies aiming to streamline their operations and improve customer satisfaction. By offering a centralized platform for order management, the study demonstrates how technology can drive business success and set new industry standards.

For the Owner of the Company: The owner of TechWorkz will benefit from a comprehensive system that provides better oversight of business operations. With the ability to track orders, monitor sales, and verify payments, the owner can make more informed decisions about business strategy and resource allocation. The system's automation and reporting features also reduce the manual workload, allowing the owner to focus on business growth and development.

For the IT Administrator: The IT administrator plays a critical role in maintaining and optimizing the web-based system. This system will provide IT administrators with powerful tools to ensure smooth operations, including system monitoring, troubleshooting, and updating functionalities. The study highlights how the IT administrator can use these tools to prevent system downtimes, secure sensitive client and company data, and maintain high performance. The system's modular design also allows IT administrators to implement future updates with ease, ensuring scalability and adaptability to changing business

For the Staff of the Company: The staff will find their work streamlined through the system's centralized platform for managing client inquiries, tracking orders, and monitoring 3D printing progress. This improved workflow will allow staff to be more efficient, reducing the need for repetitive tasks and manual record-keeping. The chat module will enable better

communication with clients, while the reporting features will help staff assess their performance and improve customer satisfaction.

For the Clients: Clients of TechWorkz will experience a significant improvement in their interaction with the company. The new system will offer a more user-friendly and efficient way to access information, place orders, and track their 3D printing projects. Clients can also benefit from accurate cost estimation, transparent payment monitoring, and real-time order tracking. These features will enhance the customer experience and foster greater trust in the company's services.

For the Guests: Even individuals who are not registered clients of TechWorkz can benefit from the new system. Guests can access essential information about the company's services, including frequently asked questions (FAQs), sample galleries, and ready-to-print 3D models, without needing to create an account. This allows them to explore what TechWorkz offers before committing to an order. Additionally, the system provides a contact form for guests to send inquiries, ensuring a straightforward way to communicate with the company. This open accessibility can attract new clients and expand the company's customer base.

For the IT Industry: The broader IT industry benefits from this study as it demonstrates the importance of web-based systems in enhancing business operations. By providing a successful example of technology-driven solutions, the study contributes to the ongoing development and innovation in the IT sector. It highlights the value of integrating technology into traditional business processes to improve efficiency, customer satisfaction, and overall business success.

Scope and Limitation

The "Web-Based 3D Printing Service Ordering Management System for TechWorkz" is designed to address a range of operational challenges faced by the company and to enhance the efficiency of its 3D printing service ordering process. The primary objective of this system is to provide a seamless and centralized platform that integrates various functionalities to streamline the entire process from initial client inquiry to final product delivery. One of the core features of the system is user account management, which will enable clients to create and manage their accounts securely. This functionality will support secure logins, personalized dashboards, and access to order histories. By providing clients with a user-friendly interface to handle their orders, the system will significantly reduce the administrative burden on Company's staff.

Order processing is a critical component of the system. Clients will be able to submit their orders online, which includes uploading custom 3D models they wish to be printed. The system will facilitate detailed cost estimation, taking into account factors such as material usage, printing time, and the complexity of the designs. This will provide clients with transparent pricing and help them make informed decisions before placing their orders. Real-time order tracking will be integrated into the system, allowing clients to monitor the progress of their orders from the initial stages through to completion and delivery. This feature will improve client satisfaction by providing visibility into the status of their orders and helping them manage expectations.

Additionally, the system will feature a comprehensive photo gallery showcasing previous 3D-printed works, providing clients with visual references and demonstrating the quality and variety of TechWorkz's offerings. A catalog of ready-to-print models will also be available,

offering clients a selection of pre-designed models to choose from. This will cater to clients who may not have custom designs but wish to utilize the 3D printing services. To enhance communication between clients and staff, the system will include a chat feature, formalizing interactions and reducing reliance on informal channels like Facebook Messenger. This will enable clients to ask questions, request updates, and receive timely responses, thereby improving overall customer service.

Payment tracking is another essential feature of the system. Clients will be able to upload proof of payment via email, which staff will verify and update. Gcash and Paymaya payment options will streamline the payment process, reduce the potential for errors, and ensure that payment statuses are accurately recorded. The system will also generate various reports, including lists of client orders, sales reports, and summaries of completed orders. These reports will provide valuable insights into business performance, helping the Company's make informed decisions and manage operations more effectively.

Despite these comprehensive features, the system has several limitations that must be acknowledged. One significant limitation is the lack of direct payment processing capabilities. The system will not handle payment transactions directly, but using Paymongo as payment gateway test payment processor only for the system demo. Additionally, the system is not designed to manage physical logistics; while it will track the status of 3D printing orders, it will not handle the actual shipping or delivery of the final products. This means that logistics and coordination with delivery services will need to be managed separately, which could affect the timeliness of order fulfillment.

Performance issues may also arise due to external factors. The system's availability and responsiveness could be impacted by factors such as internet connectivity and server

stability. Any technical problems with the hosting provider or internet outages could affect the system's performance, potentially leading to disruptions in service. Furthermore, the system does not integrate with other business software, such as accounting for inventory management systems. This necessitates separate management of these aspects, which could lead to inefficiencies and potential data discrepancies. Integration with existing business software could enhance the system's functionality and streamline overall operations, but this will not be included in the initial implementation.

Development challenges are also a consideration. Technical issues such as bugs or security vulnerabilities may arise during development or after implementation, requiring ongoing maintenance and support. Additionally, user adoption and training will be essential to ensure that both staff and clients can effectively use the new system. This will involve allocating time and resources to familiarize users with the system's features and functionalities. Effective communication between developers, stakeholders, and users will be crucial to identifying and addressing potential issues early. Regular testing and feedback sessions can help ensure the system aligns with user needs and expectations. Furthermore, scalability should be considered to accommodate future growth or changes in demand. can set realistic expectations and work towards a successful implementation of the "Web-Based 3D Printing Service Ordering Management System.

Definition of Terms

The proposed system involves various technical and operational terms that require clear definitions. This section ensures that readers have a consistent understanding of the language used throughout the study. The technical terms relate to the technological aspects

and features of the system, while the operational terms focus on how these concepts function within the specific context of the project.

Technical Terms

3D Printing. The process of creating three-dimensional objects by adding material layer by layer based on a digital model. This additive manufacturing technique allows for the production of complex and customized objects.

Computer-Aided Design (CAD). A technology used to create, modify, analyze, or optimize designs in a digital environment. CAD software is commonly used to design 3D models for printing, enabling precise control over shapes and dimensions.

Database Management System (DBMS). A sophisticated software application designed to manage, store, and organize data efficiently. In the context of a web-based 3D printing system, the DBMS handles client information, order specifics, payment records, and supports reporting and analysis.

Extrusion-Based Printing. A common 3D printing technology where material, usually thermoplastic, is heated and extruded through a nozzle to build the object layer by layer. This method is popular due to its versatility and affordability.

Material Filament. The raw material used in 3D printing, typically in the form of a long spool of thermoplastic. Filaments come in various types, such as PLA (polylactic acid) and ABS (acrylonitrile butadiene styrene).

Post-Processing. The steps taken after a 3D object is printed to improve its finish, remove support structures, or add coatings for durability. Post-processing can involve sanding, polishing, or painting.

Quality Control. The process of ensuring that a 3D-printed object meets the desired specifications and quality standards. This involves checking for defects, measuring dimensions, and testing durability.

Resin-Based Printing. A 3D printing technology that uses liquid resin cured by ultraviolet light to create objects. This technique is known for producing high-resolution prints with excellent details.

Slicing Software. Specialized software used to convert a 3D digital model into thin horizontal layers, which are then used by the 3D printer to build the object. Slicing software determines the accuracy and quality of the final print.

Web Hosting. A service that allows individuals or organizations to host a website on the internet. In the context of a web-based 3D printing service, web hosting enables the platform to be accessible to clients online.

Operational Terms

Chat System. is a communication platform within the system that enables real-time messaging between clients and staff. This term includes all interactions for customer support, order inquiries, and general communication.

Client. refers to any individual or organization that creates an account within the system to access 3D printing services.

Dashboard. is the main user interface within the system that provides users with a comprehensive view of the available features and functionalities. The dashboard is where clients and staff can navigate through the system to perform various tasks, such as placing orders, managing accounts, and tracking order status.

Guest. a user who accesses the web-based system without creating a registered account. Guests are allowed to view publicly available content, such as product galleries and frequently asked questions (FAQs).

Order. is the formal request initiated by a client within the system for a specific 3D printing service. This term encompasses all aspects of the transaction, including the type of product, custom specifications, and payment details.

Payment. denotes the financial transaction made by clients to complete their orders. This term includes a variety of payment methods, such as online bank transfers, mobile payment platforms, and credit card transactions.

Portal Direct Entry. refers to the interface through which users access the system by entering a specific URL into a web browser.

Progress Status. refers to the updates provided to clients regarding the current stage of their 3D printing orders. This term encompasses a range of information, including estimated completion times, any delays or issues, and final delivery details.

Quotation. refers to the detailed estimate provided by the system or staff, indicating the expected cost and terms for a specific 3D printing service. This term encompasses the breakdown of costs based on factors such as materials, complexity, and time required for production.

Quality Check. the final production stage ensures 3D printed products meet quality standards and customer specifications before delivery. It involves thorough inspections to identify and address defects, ensuring products are accurate, functional, and of superior quality.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter groups the related studies by keywords to highlight common themes and insights relevant to the proposed Web-based 3D Printing Service Ordering Management System for TechWorkz. The keywords indicate primary areas of focus, including additive manufacturing, cloud computing, supply chain management, and e-commerce.

The study presents a comprehensive analysis of the innovative application of additive manufacturing (AM) techniques within the domain of magnetic materials. It specifically delves into the utilization of laser engineering net shaping (LENS) and other advanced methods to fabricate both soft and hard magnets. This investigation reveals the substantial impact of AM on the production of magnetic components, particularly when dealing with intricate and sophisticated geometries that traditional manufacturing approaches struggle to achieve. By employing LENS and similar technologies, the study demonstrates how AM can overcome the constraints associated with conventional methods, thereby allowing for the creation of magnetic materials with highly detailed and customized structures. The authors highlight the significant advantages of using AM for developing magnetic components. One of the key benefits is the ability to produce complex geometries that are often challenging or impossible to achieve through traditional fabrication processes. The study underscores the potential of AM to revolutionize the field by enabling the production of magnetic materials with unique properties and configurations tailored to specific applications. Furthermore, the research explores various techniques within the AM framework that contribute to the advancement of magnetic material fabrication, showcasing the flexibility and adaptability of AM technologies. The research emphasizes that AM's capacity to generate complex magnetic structures is not

limited to a specific type of magnet but spans across both soft and hard magnetic materials. This flexibility makes AM a valuable tool in the development of advanced magnetic components for diverse industrial and technological applications. The findings suggest that the integration of AM into the manufacturing process can lead to improved performance and efficiency, offering new opportunities for innovation in the production of magnetic materials. Overall, the research highlights the transformative potential of additive manufacturing in the field of magnetic materials. By providing a detailed analysis of LENS and other AM techniques, the study contributes valuable insights into how these technologies can be harnessed to produce high-quality magnetic components with complex geometries, paving the way for future advancements in the field (Chaudhary et al., 2020)

Metal additive manufacturing (AM) represents a dynamic and rapidly advancing sector, characterized by cutting-edge technologies and processes. This review offers an in-depth examination of current research on metal AM, detailing its capabilities and constraints. It provides an overview of fundamental principles, including the intricate physics of melt pool formation, and addresses prevalent defects associated with this technology. The availability of metals and alloys for commercial use is discussed, alongside methods for process optimization, computational modeling, and post-processing techniques aimed at enhancing surface quality, mechanical strength, and dimensional precision. Although the variety of printable alloys is expanding, further development beyond existing commercial options is necessary. The lack of a comprehensive numerical model for AM, largely due to computational limitations, remains a challenge. Despite this, ongoing advancements indicate that metal AM is poised for broader adoption. Additive manufacturing uses a 3D model to build objects layer by layer, enabling the creation of complex geometries that traditional methods cannot achieve. This approach facilitates the production of intricate internal lattice

structures, which offer strength while minimizing weight by optimizing material distribution. AM also reduces manufacturing time and material waste by allowing for on-demand production and provides new possibilities for controlling and adjusting the microstructure of printed items, thus affecting their mechanical properties. However, the range of metal alloys that can be effectively printed remains limited compared to conventional methods. The printing process involves phase transformations due to layer deposition and rapid cooling, which vary with equipment, geometry, and parameters. These variations can cause defects like long columnar grains, porosity, and cracks, impacting the material's mechanical performance. Despite these complexities, AM-manufactured alloys often show finer microstructures and higher strengths compared to those made through traditional methods. The Wohlers Report 2019 highlights a 41.9% increase in revenue from metal AM, continuing a trend of substantial annual growth. Industries such as automotive and aerospace are increasingly leveraging the advantages of metal AM. As the field evolves, understanding the interaction between material properties, microstructure, and processing techniques is crucial for producing high-quality parts suitable for engineering. While there are many reviews on metal AM, they are often outdated or focused on specific issues like defects or optimization. This review provides a holistic view of metal AM by discussing prevalent processing methods, the physics behind them, and the impact of process parameters on defects. It also covers the current state of processable alloys, optimization techniques, computational modeling, and post-processing strategies. (Cooke, S et al., .2020).

E-commerce has rapidly evolved into a crucial facet of the global economy, with its significance dramatically heightened during the COVID-19 pandemic. The surge in online shopping has necessitated a deeper understanding of performance metrics and forecasting to help businesses navigate and excel in this competitive landscape. This paper introduces

a sophisticated hybrid approach to evaluate and predict e-commerce performance by integrating the Grey model (GM (1, 1)) with Data Envelopment Analysis (DEA), specifically the Malmquist-I-C model. The GM (1, 1) model is utilized to forecast future performance indicators for the e-commerce sector from 2020 to 2022. This predictive model is essential for anticipating future trends and guiding strategic decisions. Concurrently, the Malmquist-I-C model assesses the efficiency and productivity of top e-commerce companies by analyzing input variables such as assets, liabilities, and equity, alongside output variables like revenue and gross profit. The application of these models to the top 10 e-commerce companies in the U.S. provides a comprehensive evaluation of their performance. The results reveal that eBay has been the most productive e-commerce company on average over the study period, followed by Best Buy and Lowe's. Conversely, Groupon was identified as the least efficient performer. This analysis underscores the pivotal role of technological advancements in driving productivity growth. Despite the significant focus on technological development, the study highlights that maximizing resource utilization—including labor, materials, equipment, and capital—is also critical for enhancing efficiency. This research offers valuable insights for decision-makers in the e-commerce industry. By combining GM (1, 1) and DEA, it provides a robust framework for assessing historical, current, and future performance. This approach enables businesses to identify key success factors and make informed strategic improvements. The study emphasizes the importance of leveraging both predictive and evaluative tools to navigate the complexities of the e-commerce market and achieve sustainable growth. The paper is structured to first review relevant studies on e-commerce performance and technological advancements. It then discusses the methodologies of the Grey model and DEA, followed by an analysis of the results. The final sections cover the discussion, conclusions, and limitations of the study. This integrated approach aims to

support e-commerce businesses in enhancing their decision-making processes and achieving long-term success in an increasingly digitized world. (Chia-Nan Wang et al., 2020).

The paper provides a comprehensive review of the application of Fused Deposition Modeling (FDM) in the pharmaceutical industry, specifically focusing on its use for creating single- and multi-compartment hollow systems designed for oral drug delivery. FDM is a 3D printing technique that allows for the layer-by-layer deposition of material to build up complex structures from a digital model. This review highlights the feasibility of FDM in manufacturing hollow drug delivery systems that can effectively convey various formulations for oral administration. The review begins by discussing the advantages of FDM in pharmaceutical applications, particularly its ability to separately develop the inner core and outer shell of a drug delivery system. This separation allows for significant flexibility in designing systems that can control the release of the drug. For example, the outer shell can act as a barrier to control the release rate of the drug from the inner core. The paper discusses both systems that are composed of parts which are filled and assembled post-fabrication and those that are fabricated and filled in a single manufacturing process. The review covers various configurations of these systems, starting from simple, single-compartment designs, which resemble traditional capsules, and extending to more complex multi-compartment systems. These advanced systems can be filled with different formulations, left empty to ensure floatation, or designed to achieve combined release kinetics. Each system's design, formulation approaches, manufacturing processes, and release performances are critically analyzed. The versatility of FDM is underscored, particularly in terms of the geometric freedom it provides, which allows for the creation of both simple and intricate geometries in drug delivery systems. The paper also addresses some of the limitations associated with FDM, such as production speed, cost, and labor, which are significant challenges for broader

adoption in the pharmaceutical field. The potential of FDM in personalized medicine is explored, noting that it enables the on-demand and decentralized production of drug products tailored to individual patient needs. This capability could lead to significant improvements in patient compliance and cost-effectiveness, particularly for those with rare diseases or complex medication regimens. Moreover, FDM could support telemedicine by enabling the remote manufacturing of personalized medicines based on virtual prescriptions. (Alice Melocchi et al., 2020)

The evolution of 3D printing technology has significantly impacted various sectors, including industry, healthcare, sports, and education. With rapid advancements in technology and the continuous development of new materials, 3D printing has become increasingly prevalent. The global 3D printing market has seen substantial growth, with annual growth rates reaching 24.9% from 2013 to 2017 and a total industry value of USD 7.34 billion in 2017. This growth underscores the need for effective management of 3D print resources, particularly in a distributed network manufacturing environment. To address the challenges posed by the expansion and diversity of 3D printing devices, this study introduces a comprehensive approach based on cloud manufacturing principles. The proposed method involves a dynamic and static data-based matching system designed to optimize resource management for cloud 3D printing. This innovative approach utilizes a modeling framework that integrates model-based systems engineering (MBSE) to describe and manage both print tasks and print resources. A key feature of this approach is the development of a universal dynamic data acquisition system for 3D printers. This system enables real-time monitoring and automatic diagnosis of printer health, addressing the limitations of traditional static models. By incorporating both static characteristics, such as printer parameters, and dynamic characteristics, such as real-time operational data, the system provides a more

comprehensive and responsive solution for managing 3D print resources. The study highlights the importance of effectively matching print tasks with available resources in a cloud-based environment. The proposed dynamic and static data-based matching method is designed to simulate the supply-demand matching process accurately, offering a technical solution for efficient resource allocation on a large scale. This approach not only enhances the accuracy of resource matching but also improves the overall efficiency of cloud manufacturing operations. By integrating real-time data and addressing the dynamic nature of 3D printing resources, the study presents a significant advancement in managing 3D print resources within a cloud manufacturing framework. This method overcomes the challenges associated with traditional static models and provides a robust solution for optimizing resource management in the evolving landscape of 3D printing technology. (Xiao Luo et al., 2019)

Additive Manufacturing (AM) represents a significant advancement in the field of manufacturing and production, marking a shift towards the digitization of these processes. This innovative technology is based on the principle of building up products layer by layer from three-dimensional digital designs, which are meticulously crafted using sophisticated software and hardware. The method involves precisely depositing materials to create complex and customized products directly from digital models. This approach offers a multitude of benefits, including notable cost efficiency, a reduction in time-to-market, increased flexibility in design, and the ability to produce highly customized items tailored to specific needs. These advantages make additive manufacturing particularly appealing for various industries that require rapid prototyping and bespoke solutions. Despite its numerous benefits, additive manufacturing is not without its challenges. Among the primary issues are ensuring reliable traceability of products throughout the manufacturing process, obtaining

certification to guarantee quality compliance, and protecting intellectual property from potential misuse or theft. These challenges can hinder the broader adoption and trust in additive manufacturing technologies. To address these challenges, the integration of blockchain technology offers a promising solution. Blockchain's distributed ledger system provides a robust framework for enhancing product traceability and building trust among all participants in the manufacturing supply chain. By leveraging blockchain, manufacturers can ensure that every transaction is securely recorded and immutable, which greatly improves transparency and accountability. The paper proposes a blockchain-based solution specifically designed to tackle the traceability issues associated with additive manufacturing. This solution aims to guarantee secure, trusted traceability, accessibility, and immutability of transactions and data provenance among all stakeholders involved in the supply chain. The proposed system utilizes Ethereum smart contracts to manage and track transactions initiated by participants in the manufacturing process, ensuring that all actions are recorded and validated. Additionally, it employs decentralized storage through the Inter-Planetary File Systems (IPFS) to securely store and share critical data such as design files, records from Internet of Things (IoT) devices, and other product specifications. The paper provides a comprehensive overview of the system architecture, detailing its major components, key stakeholders, and interactions within the framework. It includes a thorough description of the implementation process and the algorithms that underpin the operation of the proposed solution. Furthermore, the paper presents an in-depth analysis of the security and cost aspects of the solution, highlighting its efficiency in meeting essential security and performance criteria. By incorporating blockchain technology, the proposed solution not only addresses the existing challenges in additive manufacturing but also enhances the overall security, efficiency, and reliability of the manufacturing process. This integration represents

a significant step forward in ensuring that additive manufacturing can meet the growing demands for customized and high-quality products while maintaining rigorous standards of traceability and data integrity. (Mohammed Omar et al., 2020)

This study explores the impact of 3D printing on a straightforward supply chain involving a manufacturer and retailer responding to stochastic customer demand. Despite the increasing interest in 3D printing, a relatively new technology, its effects on operational and supply chain dynamics are not thoroughly understood. A unique aspect of 3D printing is its potential for installation at the retailer's location, enhancing the supply chain's responsiveness to demand. The research investigates the adoption of 3D printing by either the manufacturer or the retailer and evaluates equilibrium conditions in Stackelberg games for both scenarios. The findings reveal that incorporating 3D printing can lead to greater profits compared to traditional systems. For manufacturers, placing 3D printers at the retailer's site often results in the most favorable profit outcomes, though the retailer's preference can vary based on specific factors. Supply chain managers should carefully consider the potential benefits of 3D printing in their operations. 3D printing, also known as additive manufacturing, constructs products layer by layer, contrasting with traditional methods such as milling or forging. It has attracted significant attention from both governments and businesses, including major investments from companies like GE and Mercedes Benz. The primary benefit of 3D printing is its ability to produce items closer to the final customer, thereby reducing transportation and inventory holding costs. However, 3D printing generally has lower production speeds and requires substantial investment in new technology and designs. The research examines two main scenarios: one where the manufacturer invests in 3D printers, potentially located at the retailer's site, and another where the retailer adopts 3D printing technology. In the first scenario, the manufacturer

covers all 3D printing costs under a lump-sum contract with the retailer. In the second scenario, the retailer invests in 3D printing with a dual wholesale-price contract in place. These scenarios are compared to traditional supply chain models to highlight the trade-offs and advantages associated with 3D printing. This work also relates to broader discussions on supply chain management and technology investments, extending existing research by focusing on decentralized supply chains and collaborative investment strategies between manufacturers and retailers. It contributes to understanding how 3D printing influences cash, material, and information flows within supply chains. (Arbabian et al., 2020)

The emergence of 3D printing, or Additive Manufacturing (AM), marks a revolutionary shift in the manufacturing landscape by enabling the creation of complex geometries and enhancing design capabilities in ways that traditional manufacturing processes cannot. Unlike conventional methods that often require extensive tooling and setup, 3D printing builds objects layer by layer from digital designs, significantly reducing the need for physical prototypes and allowing for rapid adjustments. This capability is particularly transformative for supply chain management, offering substantial advantages such as reduced inventory levels, lower shipping costs, and decreased capital expenditures on factories and warehouses. By fabricating products closer to the point of demand, 3D printing minimizes the traditional supply chain's reliance on large inventories and long shipping times. This capability enhances flexibility, allowing for real-time customization of products to meet specific customer needs, and contributes to the reduction of waste material, as additive manufacturing processes use only the necessary amount of material for production. The various methods of additive manufacturing, including Stereolithography (SL), Fused Deposition Modeling (FDM), and Selective Laser Sintering (SLS), each bring unique advantages to different aspects of production. For example, Stereolithography enables the

creation of high-precision parts directly from CAD models without the need for tooling, while Fused Deposition Modeling is known for its cost-effectiveness in producing durable prototypes. Despite these benefits, the adoption of 3D printing in supply chains is not without its challenges. High initial costs for 3D printers and limitations in material choices can be barriers to widespread implementation. Additionally, the integration of 3D printing into existing supply chains requires careful consideration of technological and logistical factors to fully realize its potential. Nevertheless, the technology's ability to enable on-demand production, reduce lead times, and offer greater design flexibility is paving the way for a significant transformation in global supply chain management, promising enhanced efficiency and responsiveness in various industries. (M. Varsha Shree et al., 2020)

Additive manufacturing (AM), commonly known as 3D printing, has undergone rapid development, revolutionizing the way products are designed and produced. As this technology evolves, accurate cost prediction models become increasingly critical. These models play a vital role in assisting manufacturers and designers in making informed decisions during various phases of product development. The challenge of cost estimation in AM arises from the need to synchronize a vast array of manufacturing knowledge and processes, from the initial design to the final production stage. The complexity of AM cost estimation is reflected in the diverse range of models that have been developed over the years. Each model aims to address specific aspects of the cost estimation process, often tailored to particular applications or technologies within the additive manufacturing spectrum. As a result, a wide variety of cost models exist, each employing different terminologies, perspectives, and techniques. Understanding these models requires familiarity with the various approaches used in their development, as well as the contexts in which they are applied. The review of AM cost estimation models provides a comprehensive overview of the

different methods and techniques employed in the field. It highlights the evolution of these models and offers a systematic classification approach to better understand their development. The review also presents definitions of key terminologies and concepts, shedding light on the various classification techniques used in cost estimation. By examining the advancements and changes in cost estimation models over the past three decades, the review aims to provide valuable insights into the future of AM cost estimation. It emphasizes the importance of these models for different users of additive manufacturing technology, from engineers and designers to manufacturers and decision-makers. The review serves as a resource for understanding the current state of AM cost estimation and offers a glimpse into potential future developments in this critical area. (Aini Zuhra Abdul Kadir et al., 2020)

Synthesis

In the context of the proposed Web-based 3D Printing Service Ordering Management System for TechWorkz, synthesis is the process of drawing together insights from a variety of studies to form a cohesive understanding of how these elements support the system's design and objectives. It is also about recognizing the common themes, concepts, and challenges presented by these studies and examining how they relate to the specific needs of the Company. Chaudhary et al. (2020) emphasize the remarkable potential of integrating additive manufacturing (AM) into the production of magnetic materials, particularly through technologies such as Laser Engineering Net Shaping (LENS). This technology enables the precise creation of both soft and hard magnets with complex geometries, which traditional manufacturing methods struggle to achieve. For soft magnets used in transformers and electric motors, AM's capability to create intricate internal structures optimizes magnetic flux and minimizes losses. Similarly, hard magnets, essential in permanent magnet applications,

can be customized to meet specific performance requirements through AM. However, challenges such as material properties and process control remain, necessitating ongoing research to fully harness AM's capabilities in this domain.

Cooke et al. (2020) offer a comprehensive review of metal additive manufacturing, highlighting its substantial growth and potential. Techniques such as Selective Laser Melting (SLM) and Direct Metal Laser Sintering (DMLS) are noted for their ability to produce complex metal parts with minimal material waste, which is crucial in industries like aerospace, automotive, and medical implants. The review underscores the advantages of metal AM, including the creation of parts with internal lattice structures that reduce weight while maintaining strength. Nonetheless, challenges such as phase transformations during printing and limited alloy options persist. Continued advancements in computational modeling, process optimization, and material science are essential to overcome these obstacles and broaden the range of metals and alloys available for AM.

Arbaban et al. (2020) and Varsha Shree et al. (2020) explore the transformative impact of 3D printing on e-commerce and supply chain management. Their research demonstrates how AM enables the production of products closer to the point of demand, thereby reducing transportation costs, inventory levels, and lead times. This shift towards on-demand manufacturing and just-in-time inventory systems enhances operational efficiency and customer satisfaction. By implementing 3D printing at retail sites, businesses can minimize the need for large inventory holdings and decrease environmental impact. However, high initial investment costs and limitations in current printing materials present challenges to widespread adoption. Despite these hurdles, the potential for 3D printing to significantly improve supply chain efficiency and responsiveness is considerable.

Omar et al. (2020) address the integration of blockchain technology with additive manufacturing, focusing on how blockchain can resolve issues related to product traceability and intellectual property protection. The use of blockchain's distributed ledger system and smart contracts in conjunction with AM provides secure, transparent, and immutable records of transactions throughout the manufacturing process. This integration enhances trust and efficiency by offering a tamper-proof record of each step and facilitates the protection of intellectual property through secure tracking of digital designs. This technological synergy supports the growing demand for customized products while maintaining stringent standards of traceability and data integrity.

Overall, the synthesis of these studies reveals a common theme: the significant advancements in additive manufacturing and its impact on various sectors, including magnetic materials, metal manufacturing, e-commerce, and supply chain management. The research highlights the transformative potential of these technologies in improving efficiency, performance, and sustainability. However, several gaps and areas for improvement remain, particularly in the integration of online ordering platforms with 3D printing services. One major challenge is the lack of standardization in file formats, material specifications, and print quality across platforms, which often leads to inconsistencies in the final products. Additionally, user interface and experience in many online 3D printing portals are not intuitive, posing barriers to non-technical customers or small businesses that could benefit from on-demand manufacturing.

There is also a need for better cost transparency, real-time production tracking, and automated feedback mechanisms for design optimization. Many platforms lack dynamic pricing models that adjust based on material usage, complexity, or lead times. Furthermore,

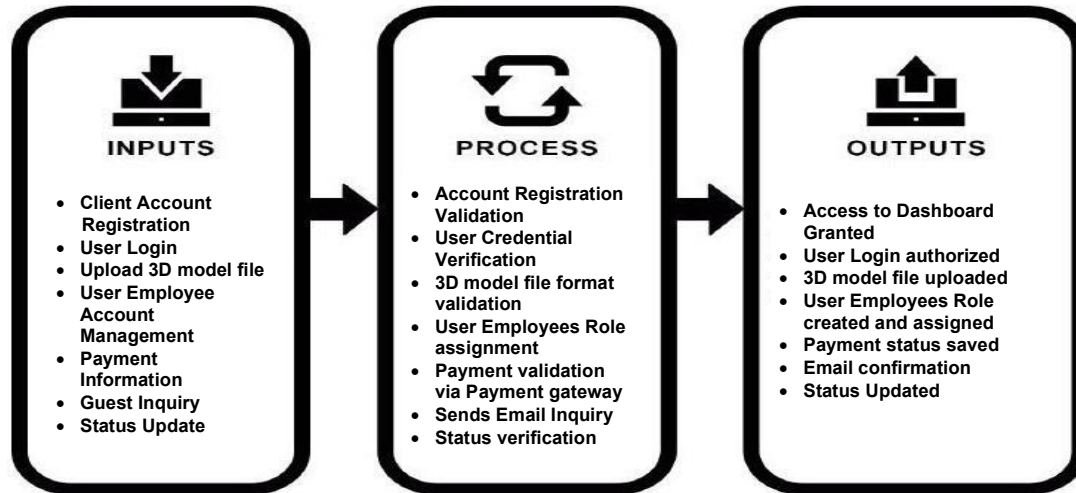
issues around intellectual property protection and secure file sharing continue to hinder broader adoption, especially for customized or proprietary designs. Addressing these challenges and leveraging the existing advancements will likely lead to continued innovations and improvements in the manufacturing and supply chain domains, shaping the future of these industries through more accessible, efficient, and customer-centered solutions.

Conceptual Framework of the Study

The conceptual framework for the proposed Web-Based 3D Printing Service Ordering Management System outlines the anticipated relationships between the various components of the system and the outcomes they aim to achieve. This framework acts as a blueprint for understanding how the different elements interact to provide a smooth and efficient experience for clients while enhancing operational efficiency for the company. By integrating a centralized platform for managing orders, payments, and inquiries, the system is designed to streamline processes and improve client satisfaction. The framework also highlights the expected improvements in communication and workflow, which are crucial for optimizing service delivery and overall business performance. Furthermore, it emphasizes the role of automation in reducing errors and processing time. The system aims to provide real-time order tracking, ensuring transparency for clients. Enhanced data management capabilities will support better decision-making and strategic planning. Overall, the framework underscores how technology can drive innovation and growth in the 3D printing industry.

Figure 2

Conceptual Framework - IPO Diagram



The conceptual framework of the proposed system, as shown in Figure 2, for the Web-Based 3D Printing Service Ordering Management System includes three core elements: input, process, and output, each critical to the system's functionality and success. Inputs are the initial components that drive the system. They include Client Orders, which provide details about the products requested, such as type, quantity, material, and deadlines. Payment Details include financial information necessary for transaction processing, such as credit card or digital wallet data, ensuring secure payments. Design Files are the 3D models or blueprints essential for the printing process, and Customer Inquiries involve communications from clients seeking information or assistance. These inputs are essential for initiating and guiding the system's operations. Processes convert these inputs into outputs through several coordinated activities. Order Management involves receiving, assessing, and organizing client orders, scheduling, and tracking resources to fulfill requests. 3D Printing involves manufacturing the products from design files using various technologies, requiring oversight to ensure quality. Payment Processing manages financial transactions, including

confirmations and refunds. Customer Communication maintains dialogue with clients, providing updates and support, while Quality Control ensures that finished products meet required standards. These processes work together to ensure efficient operation and output quality. Outputs are the results of the system's processes. Finished Products are the completed items ready for delivery. Customer Feedback provides insights into satisfaction and areas for improvement. Reports and Analysis offer data on operational efficiency and sales trends, aiding in business decisions. Improved Customer Satisfaction reflects the effectiveness of the system in meeting client needs and expectations.

Additionally, this framework promotes seamless interaction among stakeholders, including customers, operators, and managers. By integrating user-friendly interfaces, the system allows clients to conveniently place orders, upload designs, and track progress. For operators, automated order management reduces manual workload, ensuring timely delivery and consistency in services. The framework also includes data encryption measures to protect sensitive client information, enhancing trust in the platform. Regular system updates and maintenance ensure reliability and adaptability to evolving business needs. Incorporating analytics tools within the system enables detailed performance reviews and identification of areas for optimization. Scalability features allow the system to handle increased demand without compromising efficiency. Effective resource allocation within the process stage minimizes waste and maximizes productivity. This holistic approach ensures the proposed system's relevance and competitiveness in the dynamic 3D printing industry.

CHAPTER III

METHODOLOGY, RESULTS, AND DISCUSSION

Requirement Analysis

Requirements analysis is a critical initial phase in the development of the proposed system. This process involves a systematic approach to understanding and defining the specific needs and conditions that must be addressed to create an effective and user-friendly platform. As this phase lays the foundation for the project's success, it is essential to identify and reconcile the requirements of various stakeholders, including customers, service providers, and technical teams.

In the context of this project, the analysis will focus on ensuring that the system meets not only the functional requirements—such as user interface design and order processing capabilities—but also considers technical, operational, schedule, and economic feasibility. By comprehensively assessing these aspects, we can ensure that the developed system will not only satisfy user expectations but also align with business goals and constraints, paving the way for successful implementation and adoption.

Operational Feasibility

The operational feasibility of the proposed Web-based 3D Printing Service Ordering Management System is assessed through various critical aspects that contribute to its effectiveness. Firstly, the system's web-based nature ensures high accessibility, allowing users to place orders and manage requests from any device with an internet connection, whether at home, in the office, or on the go. This flexibility is vital in a fast-paced environment where convenience and efficiency are paramount. Furthermore, the design of the system

prioritizes user-friendliness, featuring an intuitive interface that simplifies the ordering process. By employing clear navigation, visually appealing layouts, and step-by-step guidance, the system caters to users with varying levels of technical expertise, encouraging greater adoption and minimizing the learning curve.

Implementation of the web-based system is straightforward, requiring no extensive hardware installations or complex configurations. Users can access the service through standard web browsers, eliminating the need for specialized software downloads or updates. This simplicity in implementation is advantageous for organizations seeking to adopt the system quickly without significant disruption to existing operations. Additionally, operational feasibility encompasses ongoing support and maintenance; the web-based system can be updated and maintained centrally, ensuring all users benefit from the latest features and security improvements without individual updates. This centralized approach enhances reliability while reducing downtime.

The proposed system is also designed to integrate seamlessly with existing operational workflows, allowing for easy data import/export and compatibility with other software tools, such as inventory management systems. This integration supports a cohesive workflow, enabling users to manage 3D printing orders within their current processes without overhauling existing systems. Scalability is another significant factor; as user demand grows, the web-based nature of the system allows for easy scalability, with additional features and increased user capacity incorporated without significant reconfiguration or downtime. Lastly, the inclusion of feedback mechanisms enables users to provide input easily, facilitating ongoing improvements and adaptations based on user experiences, thereby fostering a user-centered approach to system development and operation.

The proposed system will demonstrate strong operational feasibility through its accessibility, user-friendliness, straightforward implementation, and ability to integrate with existing processes. By focusing on these aspects, the proposed system is poised to enhance operational efficiency and user satisfaction, making it a valuable tool for both customers and service providers.

Technical Feasibility

The technical feasibility of the proposed Web-based 3D Printing Service Ordering Management System focuses on the hardware and software specifications required for its development and implementation. This evaluation ensures that the necessary technical resources are available to support the system's functionality, performance, and reliability.

Table 1

Hardware and Software Requirements to Develop the System

Category	Specification	Description
Hardware Requirements	CPU	Multi-core processor
	RAM	16GB RAM
	Storage	250GB SSD
	Network	Internet speed minimum of 100Mbps
Software Requirements	Operating System	Windows 10 or latest
	Web Server	Apache
	Internet Browser	Google Chrome or Microsoft Edge
	Database Management System	MySQL

Web Development tool	Open-source content management system (WordPress)
Payment Gateway	GCash or Maya Gateway
Programming Languages	PHP, JavaScript, HTML, CSS
Hosting Service	Cloud-based hosting and storage service - <i>smarterasp.net</i>

The proposed system will require a robust server infrastructure to effectively host the web application as listed above in Table 1. It is recommended to utilize a cloud-based server. To develop the system, it should include a multi-core processor, to efficiently handle multiple simultaneous requests. To ensure smooth operation and processing of concurrent user activities, a minimum of 16 GB of RAM is necessary. Additionally, the system should have SSD storage with at least 250 GB capacity to facilitate fast data access and retrieval, accommodating user data, order histories, and system backups. Furthermore, a high-speed internet connection with a minimum of 100 Mbps is essential to support seamless user access and data transfer.

The Web-based 3D Printing Service Ordering Management System will be developed using a combination of software technologies that ensure a robust, scalable, and user-friendly application. The primary platform for this system will be WordPress, which will serve as the content management system (CMS) to facilitate easy management of content and user interactions. WordPress offers a flexible architecture and a wide range of plugins that enhance functionality, making it an ideal choice for this project.

The back end of the system will be built using PHP, a widely used server-side scripting language known for its effectiveness in web development. PHP will enable the creation of

dynamic web pages and facilitate seamless interactions with the database, ensuring efficient data processing and retrieval.

For data management, MySQL will be employed as the database management system. MySQL is a powerful and reliable relational database management system that allows for the storage, retrieval, and management of user data, order histories, and other essential information. Its compatibility with PHP makes it a suitable choice for the back end of the application.

To ensure accessibility and reliability, the system will be hosted on a cloud hosting service. This approach offers scalability, enabling the system to accommodate varying loads of user traffic without compromising performance. Cloud hosting services, such as *smarterasp.net* service will provide the necessary infrastructure to support the application and ensure high availability and security.

By leveraging this combination of WordPress, PHP, MySQL, CSS, and cloud hosting, the proposed system will be well-equipped to deliver a seamless user experience while maintaining the flexibility needed for future enhancements and updates.

Table 2

Hardware and Software Requirements to Implement the System

Category	Specification	Description
Hardware Requirements	CPU	Dual-core processor
	RAM	8GB RAM
	Storage	250GB SSD
	Network	Internet speed minimum of 100Mbps

Software Requirements	Operating System	Windows 10 or latest
	Internet Browser	Google Chrome, Microsoft Edge, Mozilla Firefox, Safari
	Hosting Service	Cloud-based hosting and storage service - <i>smarterasp.net</i>

On the client side, as shown in Table 2, users will be able to access the system via standard devices, including desktops, laptops, tablets, or smartphones but mainly the system is recommended to be accessed via desktops and laptops. Recommended specifications for client devices include a dual-core processor or higher for optimal performance, and a minimum of 8 GB of RAM for basic functionality and browsing. The operating system should be compatible with Windows, and users should utilize up-to-date versions of web browsers, such as Google Chrome, Mozilla Firefox, Safari, or Microsoft Edge, to ensure compatibility with modern web standards. By meeting these hardware specifications, the proposed system will be positioned for efficient operation and a positive user experience.

Schedule feasibility

Schedule feasibility involves creating a realistic timeline for the development and implementation of the project at Techworkz. This process includes mapping out all necessary activities, their durations, and dependencies, typically using a Gantt chart. It refers to the practicality of completing a project within a defined timeline. It evaluates whether the proposed schedule aligns with the project's scope, resources, and constraints. This aspect of feasibility is crucial in project planning as it ensures that the deliverables can be achieved within the stipulated timeframe

without compromising quality. Key factors influencing schedule feasibility include the complexity of tasks, the availability of skilled personnel, dependencies between project activities, and potential risks or delays. A thorough schedule feasibility analysis helps in setting realistic deadlines, optimizing resource allocation, and maintaining stakeholder confidence in the project's timely completion.

Figure 3

Gantt Chart of the Study

Activity	Apr 2024	May 2024	Jun 2024	Jul 2024	Aug 2024	Sept 2024	Oct 2024	Nov-Dec 2024	Jan-July 2025	Aug 2025
Data Gathering										
Construction 1-2										
System Analysis Construction of Chapter 3										
System Development										
Testing, Simulation and Recommendation										
Results, Conclusion										
Paper Presentation Final Defense										

The Gantt chart as shown in Figure 3 helps in detailing the project phases, tracking progress, and allocating resources efficiently. By outlining these elements, schedule

feasibility ensures that the project can be completed within a reasonable timeframe and anticipates potential delays. The objective is to guarantee that all phases of the project are finished as planned and maintain adherence to the established schedule.

Economic Feasibility

Economic feasibility assesses the financial aspects of the proposed Web-based 3D Printing Service Ordering Management System to determine whether the expected benefits justify the costs involved in its development and implementation. This analysis considers various factors, including initial development costs, ongoing operational expenses, and potential revenue generation.

The initial investment required for developing the system includes expenses related to software licensing, hardware procurement, cloud hosting services, and developer salaries. Utilizing open-source platforms like WordPress can significantly reduce software licensing costs, and cloud hosting options provide flexible pricing models that allow for cost-effective scaling based on usage, helping to keep initial expenses manageable.

After the system's deployment, ongoing costs will include server maintenance, domain registration, cloud hosting fees, and periodic updates to the software and security measures. Budgeting for these recurring expenses is crucial to ensure the system remains functional and secure. By leveraging cloud services, the system can optimize costs by scaling resources according to demand, minimizing waste.

Additionally, the proposed system has significant potential for revenue generation through various channels. It can charge users for printing services, offer subscription models for frequent users, or partner with businesses for promotional printing services. Implementing

an effective marketing strategy and leveraging the unique offerings of 3D printing can attract a diverse customer base, leading to a steady stream of income.

A detailed financial analysis should project the expected return on investment over a specific period. By estimating the revenue generated from the system and comparing it to the total costs, stakeholders can assess whether the investment is justified. A positive return on investment would indicate that the economic benefits outweigh the costs, validating the project's feasibility.

The economic feasibility of the proposed system indicates a promising outlook. By carefully managing costs and capitalizing on potential revenue opportunities, the project has the potential to be financially viable and sustainable in the long run.

Cost Benefit Analysis

A systematic approach is used to evaluate the economic feasibility of a project by comparing the anticipated costs and benefits associated with it. This analytical tool helps stakeholders make informed decisions by quantifying and assessing the financial implications of a proposed initiative. In the context of the proposed system, a CBA will involve identifying all relevant costs, including development, operational, and maintenance expenses, and contrasting them with the expected benefits, such as revenue generation, improved efficiency, and enhanced user satisfaction. By providing a clear understanding of the potential return on investment, CBA assists in determining whether the project is worth pursuing and how it aligns with the organization's strategic objectives. A comparison of the existing manual system to the proposed system will be presented and analyzed to better understand its differences and the potential savings projected for 5 years.

Table 3*Cost of the Existing Manual System*

Items	Unit Price	Quantity	Year 1	Year 2	Year 3	Year 4	Year 5	Total in 5 years
Desktop PCs	P35,000	2	P70,000					P70,000
Inkjet Printer	P8,000	1	P8,000					P8,000
3D FDM printers	P27,000	5	P135,000					P135,000
Repairs and Maintenance	P12,000	5	P60,000	P60,000	P60,000	P60,000	P60,000	P300,000
Quarterly Advertisements (online, social media, flyers, and posters)	P10,000	4	P40,000	P42,000	P44,000	P46,000	P48,000	P220,000
Monthly Staff Salary	P15,000	2	P360,000	P360,000	P384,000	P384,000	P384,000	P1,872,000
Monthly 3D Artist's Salary	P20,000	1	P240,000	P240,000	P264,000	P264,000	P264,000	P1,272,000
Monthly Telecommunications Expenses (Cellphone Load and Landline)	P1,500	12	P18,000	P18,500	P19,000	P19,500	P20,000	P95,000
Monthly Office Supplies (Pen, Papers, Printer inks)	P2,000	12	P24,000	P24,200	P24,300	P24,400	P24,500	P121,400
Monthly Internet Service Fee	P2,500	1	P30,000	P30,000	P30,000	P30,000	P30,000	P150,000
Totals			P985,000	P774,700	P825,300	P827,900	P830,500	P4,243,400

Table 3 outlines the Cost of the Existing Manual System over five years for a 3D printing-related business. It includes various operational expenses, such as equipment, staff salaries, advertisements, telecommunications, and office supplies.

The total cost of the existing system over five years amounts to ₱4,243,400. Key contributors are monthly staff salaries (₱1,872,000) and the 3D artist's salary (₱1,272,000), which collectively dominate the budget. Equipment costs like 3D FDM printers (₱135,000) and desktop PCs (₱70,000) are one-time investments, while recurring expenses such as repairs and maintenance (₱60,000), advertising (₱220,000), telecommunications (₱95,000), and internet services (₱150,000) add up significantly. Some expenses were considered to increase gradually on a yearly basis. However, according to the company's policy, a salary

increase for every 2 years of tenure of employees will be expected. This breakdown highlights the high reliance on human resources and recurring costs in the manual system, suggesting room for optimization through automation or process improvements.

Table 4

Cost of the Proposed System

Items	Unit Price	Quantity	Year 1	Year 2	Year 3	Year 4	Year 5	Total in 5 years
Desktop PCs	P35,000	2	P70,000	-	-	-	-	P70,000
3D FDM printers	P27,000	5	P135,000	-	-	-	-	P135,000
Repairs and Maintenance	P12,000	5	P60,000	P60,000	P60,000	P60,000	P60,000	P300,000
Monthly Staff Salary	P15,000	2	P360,000	P360,000	P384,000	P384,000	P384,000	P1,872,000
Monthly 3D Artist's Salary	P20,000	1	P240,000	P240,000	P264,000	P264,000	P264,000	P1,272,000
Monthly Cloud Hosting Services	P3,000	12	P36,000	P36,000	P40,000	P40,000	P40,000	P192,000
Monthly Internet Service Fee	P2,500	1	P30,000	P30,000	P30,000	P30,000	P30,000	P150,000
System Development Cost	P25,000	1	P25,000	-	-	-	-	P25,000
Totals			P956,000	P726,000	P778,000	P778,000	P778,000	P4,016,000

Table 5

Cost Comparison of the Existing Manual and Proposed System

	Year 1	Year 2	Year 3	Year 4	Year 5	Total in 5 Years
Total Cost of the Existing Manual system	P985,000	P774,700	P825,300	P827,900	P830,500	P4,243,400
Total Cost of the Proposed system	P956,000	P726,000	P778,000	P778,000	P778,000	P4,016,000
TOTAL SAVINGS BENEFITS	P29,000	P48,700	P47,300	P49,900	P52,500	P227,400

Table 4 highlights a shift in operational costs for a 3D printing-related business, emphasizing system development and cloud hosting to replace or complement the manual process. Over five years, the total cost of the proposed system is ₱4,016,000, which is ₱227,400 less than the existing manual system's cost of ₱4,243,400, as shown in the comparison table (Table 5).

Despite initial setup costs like system development (₱25,000) and one-time equipment expenses such as 3D FDM printers (₱135,000) and desktop PCs (₱70,000), recurring savings emerge from cloud hosting services (₱192,000 over five years) replacing more costly manual advertisements. The annual operational costs of the proposed system are consistently lower, averaging ₱803,000 per year, compared to ₱848,680 for the existing manual system.

This cost reduction generates cumulative savings of ₱227,400 over five years. Beyond financial savings, the proposed system likely brings additional benefits, such as increased efficiency, scalability, and improved customer experience, further justifying the investment despite upfront costs. These advantages ensure the benefits of the proposed system outweigh its costs.

Data Gathering Procedure

The data gathering procedure employed for the Web-based 3D Printing Service Ordering Management System involved a comprehensive approach to ensure that all relevant information was collected effectively. This process included three primary methods: interviews, observation, and the review of existing manual forms used by the company.

Interviews were conducted with key stakeholders, including management, employees, and end-users. These discussions aimed to uncover specific requirements, pain points, and expectations regarding the current ordering process. By engaging directly with those involved in the day-to-day operations, valuable insights were gained into the challenges faced and the desired features for the new system. The interviews allowed for open-ended questions, enabling participants to elaborate on their experiences and provide

detailed feedback that would inform the system's design. To check the details of the interview, a Transcript of the Interview is attached in Appendix A.

Observation was another critical component of the data-gathering procedure. Researchers spent time in the company's operational environment to observe how the current ordering process was conducted. This hands-on approach provided a clearer understanding of workflow inefficiencies, bottlenecks, and the interactions between staff and customers. A photo of the current 3d printing environment can be observed in Appendix C. By witnessing these processes in real-time, the development team could identify areas that required improvement and potential functionalities that could streamline operations.

Lastly, the **review of manual forms used by the company** was instrumental in understanding the existing data collection and management practices. Analyzing these forms revealed how information was currently captured, processed, and stored. This review helped identify any redundancies, data entry errors, or inconsistencies in the manual processes. Additionally, it provided a baseline for the data structure needed in the new web-based system, ensuring that essential information would be seamlessly integrated and easily accessible. See Appendix B for Screenshots of different Transactional communication.

The data-gathering procedure utilized a combination of interviews, observation, and the examination of manual forms to develop a thorough understanding of the company's current practices and requirements. This multifaceted approach ensured that the final system design would address the actual needs of users and enhance the overall efficiency of the 3D printing service ordering process.

System Architecture

The System Architecture of the proposed system serves as a blueprint that outlines the structure and components of the proposed system. This architecture is essential for

ensuring that the system meets the functional and non-functional requirements of its users while providing a clear understanding of how different components will interact. It encompasses both the hardware and software elements necessary for the effective operation of the system, illustrating how data flows through various modules and how users will engage with the application.

The architecture is designed to be scalable and flexible, accommodating future enhancements and the growing demands of users. It delineates the core components of the system, including the user interface, application server, database management, and third-party integrations. Each of these components plays a critical role in delivering a seamless user experience and ensuring reliable performance.

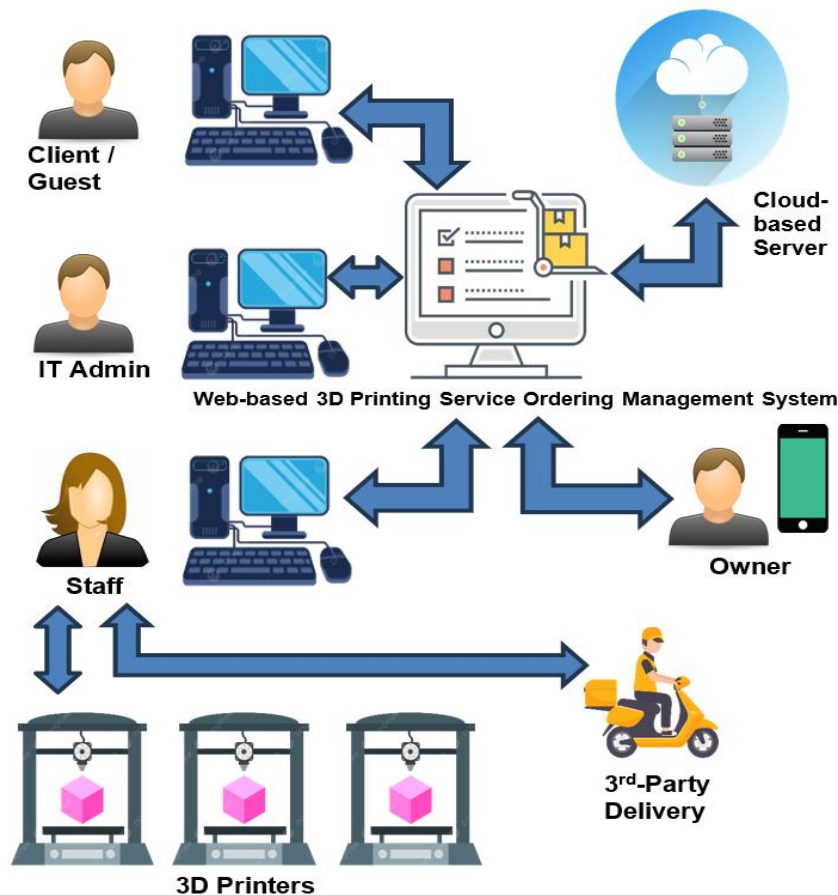
Furthermore, the system architecture emphasizes security, data integrity, and performance optimization, ensuring that user data is protected while providing fast and efficient access to services. By laying out the systematic organization of the various elements involved in the development and deployment of the system, the architecture provides a foundational framework that guides the implementation process and helps stakeholders visualize the end product. Ultimately, a well-defined system architecture is crucial for the successful realization of the proposed system, aligning technology with business objectives and user needs.

Figure 4 shows the system architecture designed to work like a typical online ordering platform, making it easy for users to upload their 3d models, place orders, and track progress. The user interface is the front-end that customers see when they visit the website or app. Through this interface, users can register or log in, upload their 3D design files, select materials, get price estimates, and place orders. The interface is designed to be simple and user-friendly so that even first-time users can navigate it easily.

Once a user places an order, the Staff takes over. It checks the uploaded file for compatibility, calculates the price, and schedules the print job. It also manages order status updates status for the users to see. This part of the system handles all the logic behind the scenes and ensures that everything runs smoothly between the customer and the service provider until the order is delivered.

Figure 4

System Architecture of the System



For the database it uses the service of a cloud server all the important information such as user profiles, uploaded design files, order details, payment history, and print job status. It keeps everything organized and accessible whenever needed.

This simple yet effective system setup makes it easy to manage 3D printing requests online, from file submission to order fulfillment, while ensuring a smooth and secure experience for both customers and administrators.

Analytical Tools

This section of the document presents the various analytical tools utilized in the analysis and design of the Web-based 3D Printing Service Ordering Management System. These tools, which include a range of diagrams, are essential for visualizing and understanding the system's functionality, user interactions, and data flow. By employing established diagramming methodologies, we can effectively communicate the system's design to stakeholders and facilitate better decision-making throughout the development process.

The Data Flow Diagram (DFD) is used to represent the flow of data within the system, starting with a context diagram that provides a high-level overview of the system and identifies the key entities or users involved. Following this, we present Level 0 and Level 1 DFDs, detailing how data is processed and the interactions between different components. Each DFD will be accompanied by an explanation of its elements, adhering to the Gane and Sarson notation for clarity and consistency.

In addition to DFDs, the Entity-Relationship Diagram (ERD) will illustrate the database structure, showing the relationships between various data entities and their attributes. This tool aids in designing an efficient and normalized database schema for the system.

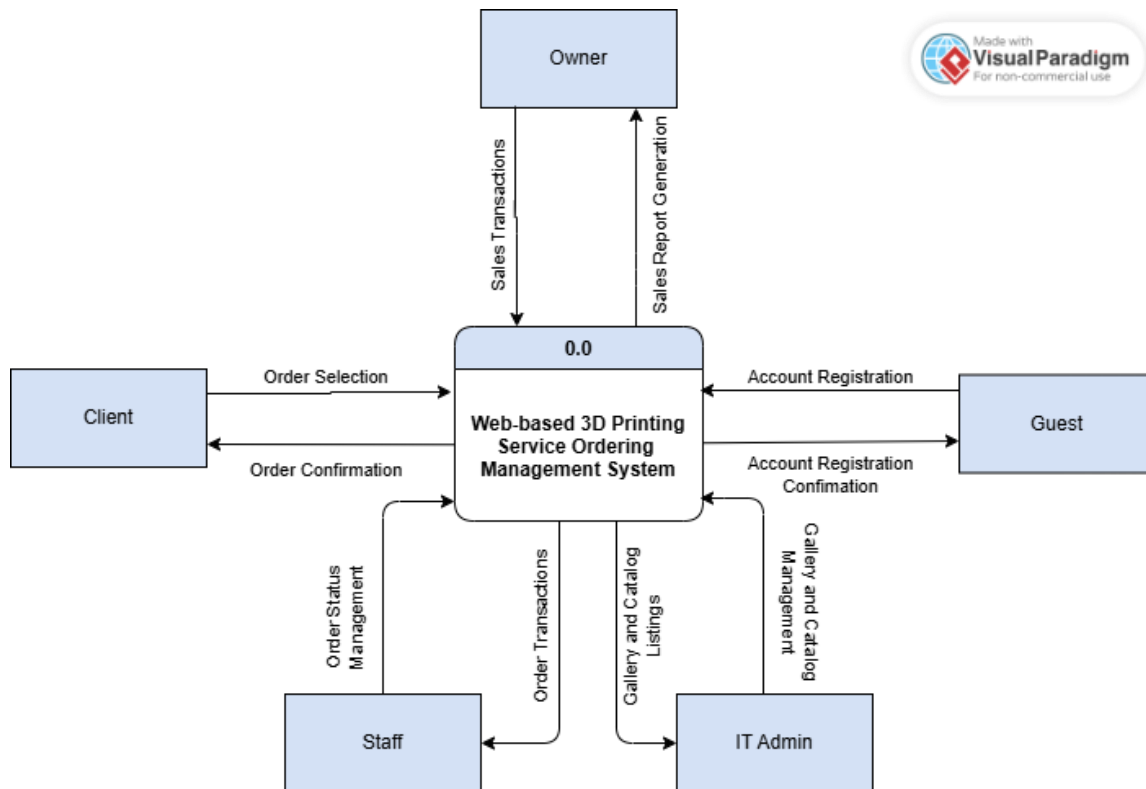
Use Case Diagrams will also be included to model the interactions between users and the system, helping identify user roles and the functionalities they can access. These

diagrams provide a clear picture of the system's use cases and the relationships between them.

Furthermore, Class Diagrams will offer a detailed view of the system's object-oriented design, showcasing the classes, their attributes, methods, and the relationships between them. These diagrams are vital for understanding the system's structure and behavior.

Figure 5

Context Diagram Level 0 of the Proposed System



The Context Diagram Level 0 of the Proposed System, as shown in Figure 5, illustrates the interactions between the main stakeholders and the Web-based 3D Printing

Service Ordering Management System. The system serves as the central hub, enabling streamlined communication and transactions between five key entities: Owner, Client, Guest, Staff, and IT Admin.

- **Clients** can browse the gallery, select orders, upload 3d model files, make payments, and receive order confirmations and status updates.
- **Guests** can send inquiries, register accounts, and receive responses, transitioning them into clients.
- **Staff** handle order processing, update transactions, and generate sales and order reports.
- **The Owner** oversees the system by receiving sales reports and managing sales transactions.
- **The IT Admin** ensures system functionality by updating gallery listings, managing user roles, and registering employee accounts.

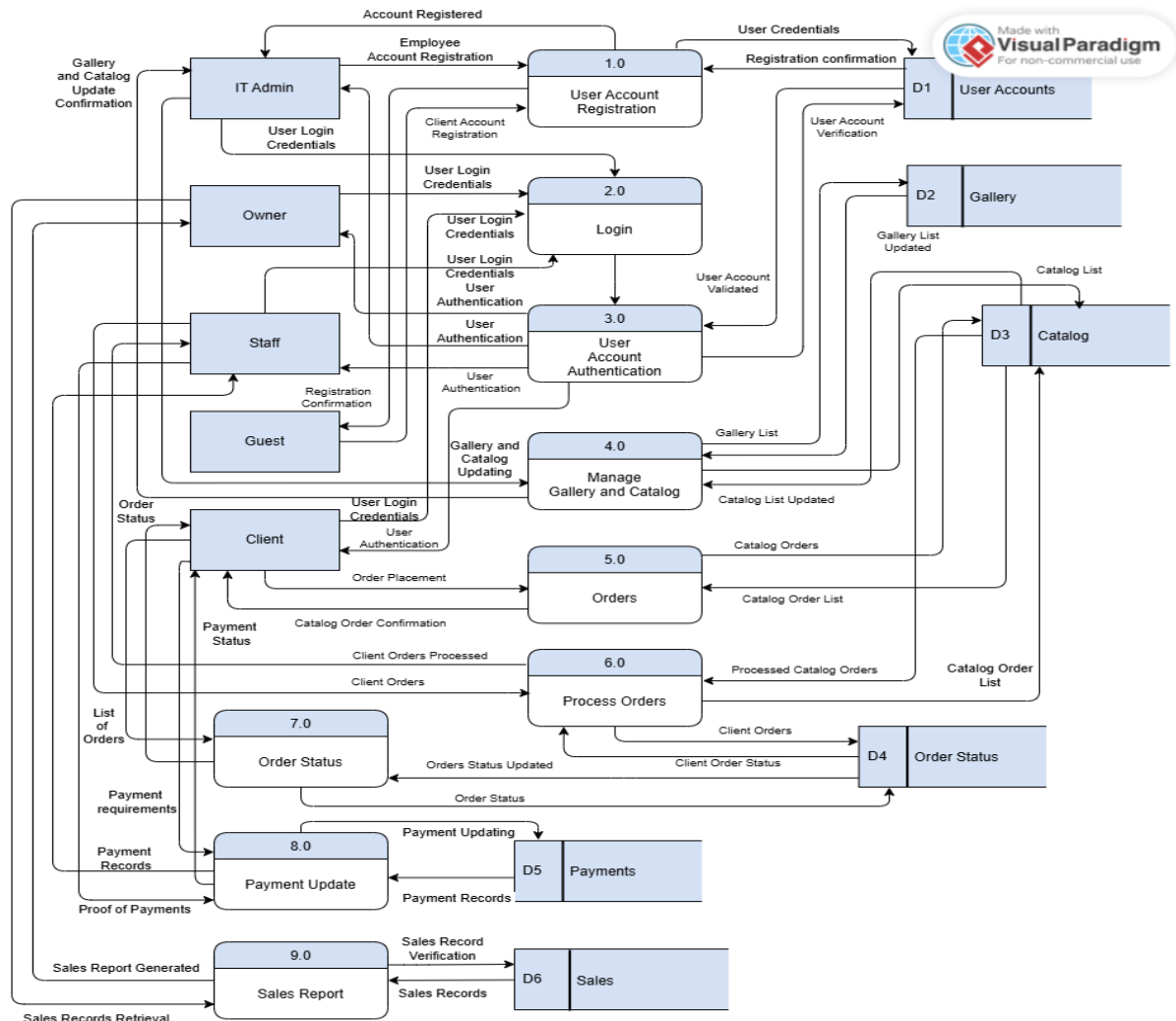
This diagram highlights the system's role in centralizing operations, improving efficiency, and enhancing user experience by automating key processes like order management, reporting, and communication.

The centralized design minimizes manual intervention, reducing errors and saving time. It facilitates real-time status updates, ensuring transparency for clients and management. By integrating reporting features, it aids decision-making for stakeholders, particularly the owner. Automated processes ensure scalability, accommodating growth in users and orders. The system also enhances efficiency by reducing human error and minimizing delays in processing. Its integration with existing infrastructure facilitates smooth transitions and operational continuity. Overall, the proposed system bridges gaps in communication and

workflow, ensuring seamless interactions among all actors. This leads to improved decision-making, faster response times, and greater flexibility in handling diverse business needs.

Figure 6

DFD Level 1 of the Proposed System



The DFD Level 1 as shown in Figure 6 depicts a comprehensive process for managing user accounts, orders, and payments. Initially, the IT Admin oversees user account registration, which leads to account creation and generates a registration confirmation. Once registered, users, including owners, staff, and guests, verify their accounts and proceed to log in via the user login process.

Different roles such as IT Admin, Owner, Staff, and Guest have specific access levels. Guests and clients, for example, can access the catalog to browse, search, and place orders. Catalog management involves updating and listing items within the system, which users can search through the catalog search feature.

Once clients place orders, they are added to the order list for processing. The system then manages these orders, preparing them for fulfillment through the order processing stage. Clients can monitor their order status, ensuring transparency and keeping them informed.

Payments are handled once orders are confirmed, streamlining financial transactions. Finally, the system generates sales reports, providing a comprehensive overview of completed sales. This organized workflow ensures efficiency and clarity throughout the entire process, from user registration to sales reporting.

Use-Case Diagram

Additionally, the Use-Case Diagram illustrates the interactions between users and the system, showcasing the various use cases in which users engage. It visually represents the relationships between the actors (users) and the functionalities of the system, providing a clear understanding of how users will interact with the web-based application, with an explanation detailing the significance of each component within the diagram.

Figure 7

Use-Case Diagram of the System

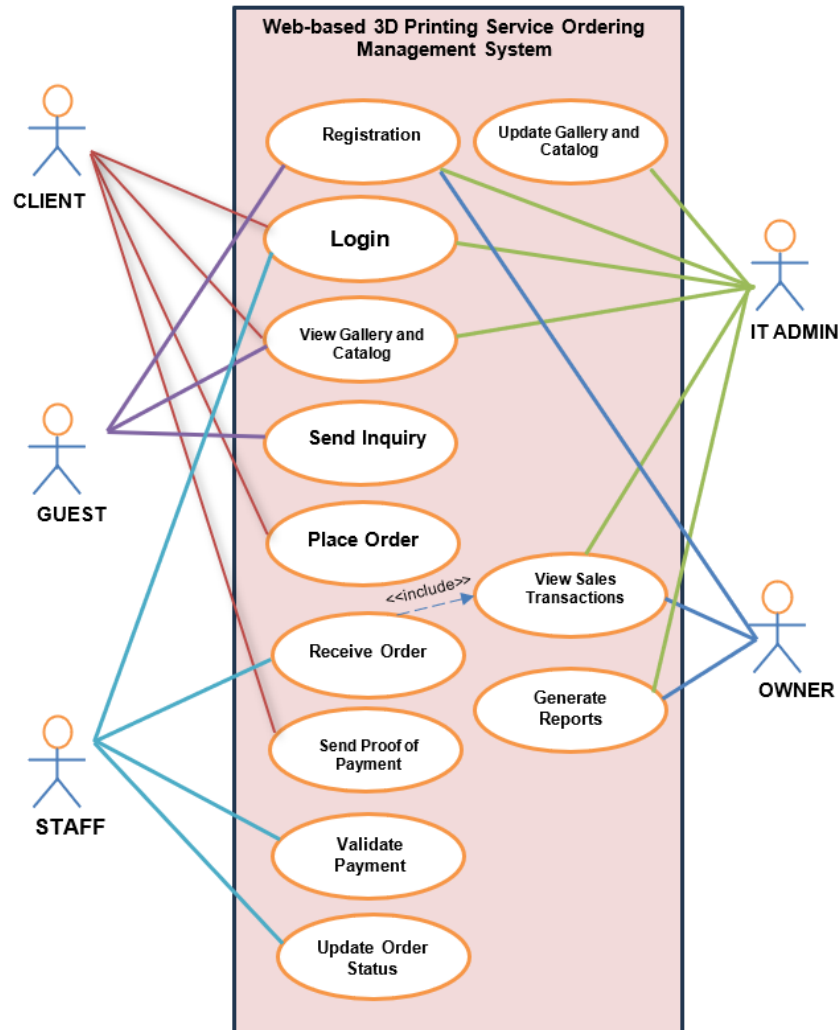


Figure 7 shows the different Users and their levels of access in the system. The lines represent User's access to each of the activities indicated. The User that has the most access to the system is the IT Admin, which involves managing the catalog and the gallery. The Owner has access to Reports generation mainly to oversee the over-all operation of the business. However, the User who has the most frequent use of the system is the Staff which accommodates Client inquiries, orders and the entire operation of the business.

Activity Diagram

Another vital component of the Unified Modeling Language (UML) is the Activity Diagram, which depicts the system's workflow and transitions between various activities.

Figure 8

Activity Diagram of the System using Swimlane

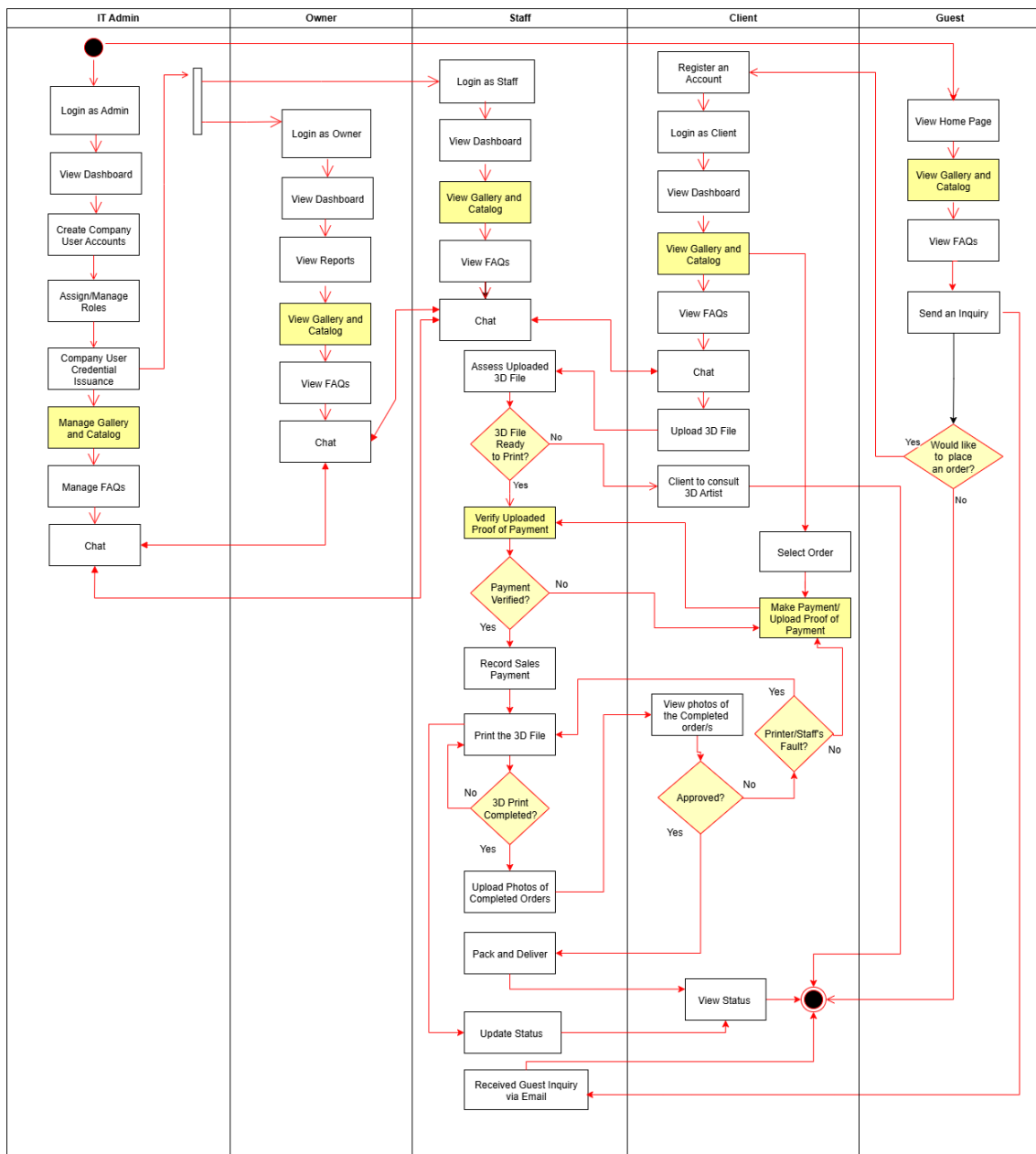


Figure 8 shows the Detailed Activity Diagram of the system using the UML Swimlane format. It represents each Entities or Users specifically of what activities are available to process, transact and interact with other Users. It gives highlight to the proposed business process flow once the system is implemented. The main transaction shows how Guest have two options to choose whether to just browse the dashboard viewing the Gallery and Catalog and send an email inquiry within the system and if the Guest got interested in placing an order an account must be created by signing up creating a Client Account that improves and gains access to other features of the system such as uploading a 3D file for quotation and assessment by the Staff. Client can also directly choose from the Catalog available ready-to-print 3D models and process payment and wait for the Staff to update the status until it manages to complete the 3D printing and becomes ready for delivery or shipment. The Owner views at the decision-maker level where generating reports is important to over-see the operational transaction of the business. While the Staff serves as the frontliner Sales Representative and operator of the 3D printing service. Staff is responsible to accommodate Client's request and orders from the Catalog making sure the smooth transaction is prioritized and giving updates real-time to the Client. The activity diagram helps the researcher to look as a reference which can be helpful in the development of the User Interface and Functionality of the proposed system.

Entity Relationship Diagram

Finally, the Entity Relationship Diagram (ERD) represents the database design of the system, showing the relationships between different data entities. The ERD will be presented in a tabular format to clarify how data is structured within the system and ensure efficient database management.

Figure 9

Entity Relationship Diagram of the System

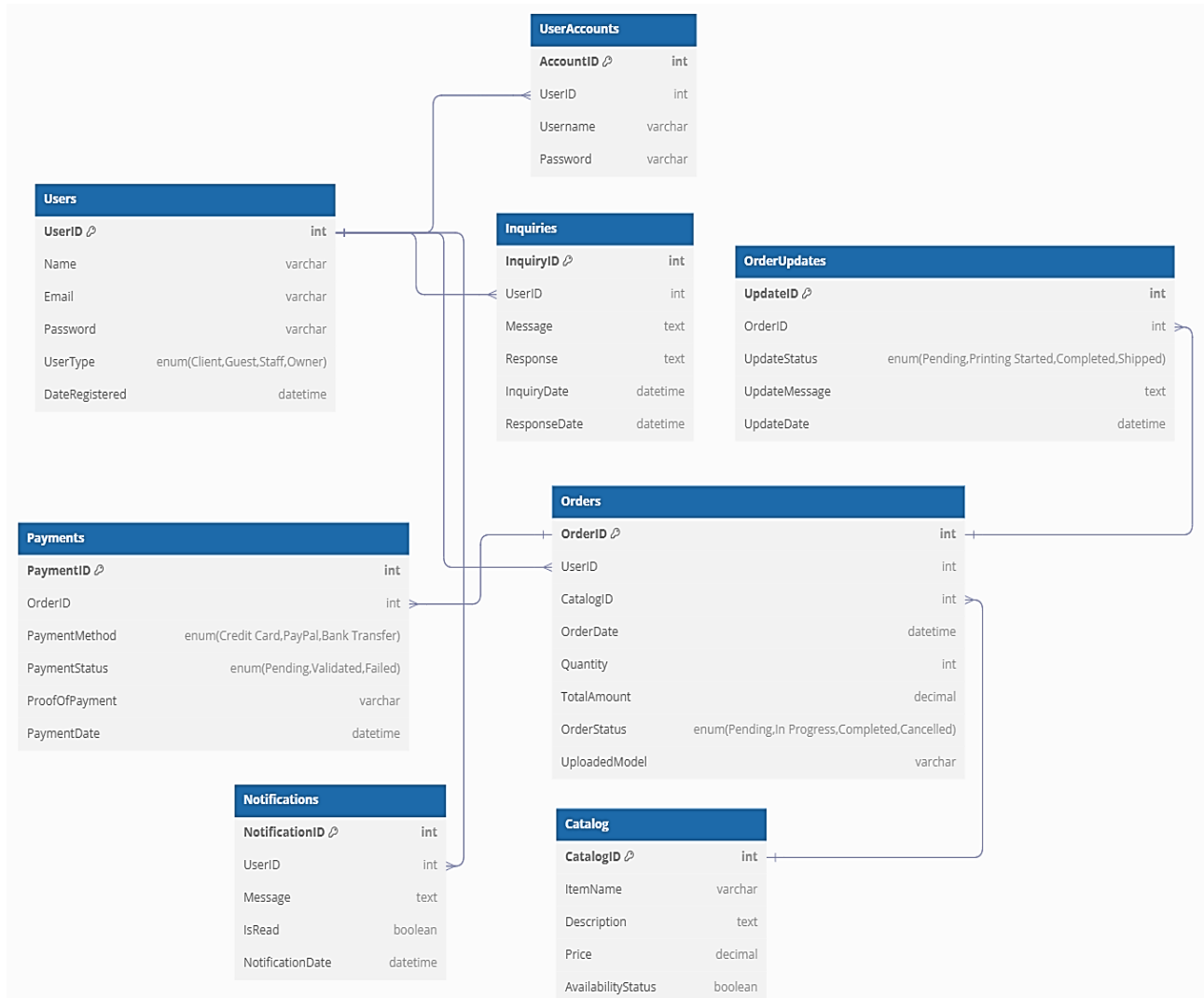


Figure 9 consists of several interconnected tables designed to manage users, orders, inquiries, payments, notifications, and catalog items.

The UserAccounts table includes fields such as AccountID (primary key), UserID (foreign key), Username, and Password. The Users table features UserID (primary key), Name, Password, UserType (which can be Client, Guest, Staff, or Owner), and DateRegistered.

Inquiries are managed in the Inquiries table, which contains InquiryID (primary key), UserID (foreign key), Subject, Message, and InquiryDate. The OrderUpdates table tracks updates related to orders with fields such as UpdateID (primary key), OrderID (foreign key), UpdateStatus (including options like Pending, Printing, Started, Completed, and Shipped), UpdateMessage, and UpdateDate.

Orders themselves are recorded in the Orders table, which has OrderID (primary key), CatalogID (foreign key), UserID (foreign key), OrderDate, Quantity, OrderStatus (with options like Pending, In Progress, Completed, and Cancelled), and UpdatedModel. The Payments table includes PaymentID (primary key), OrderID (foreign key), Amount, PaymentType (such as Gcash and Paymaya), PaymentStatus (covering Pending, Completed, Failed, and Refunded), ProofOfPayment, and PaymentDate.

Notifications for users are stored in the Notifications table, which consists of NotificationID (primary key), UserID (foreign key), Message, IsRead (boolean), and NotificationDate. Lastly, the Catalog table details items with CatalogID (primary key), ItemName, Description, Price, and Availability Status (boolean). This relational structure allows efficient management and retrieval of data across the various functionalities of the system.

Through these analytical tools, we aim to create a comprehensive visual representation of the system's design, enhancing understanding and guiding the development process to align with user needs and project objectives.

Development and Testing

The Development and Testing phase is a crucial segment of the software development lifecycle (SDLC), encompassing a structured approach to creating, implementing, and evaluating the Web-based 3D Printing Service Ordering Management System. This phase is instrumental in ensuring that the final product not only meets the defined requirements but also performs efficiently in a real-world environment.

Together, the Development and Testing phase, guided by the SDLC and the Software Evaluation Plan, ensures a robust, reliable, and user-friendly system that effectively addresses the needs of its stakeholders while maintaining high-quality standards throughout its lifecycle.

Software Development

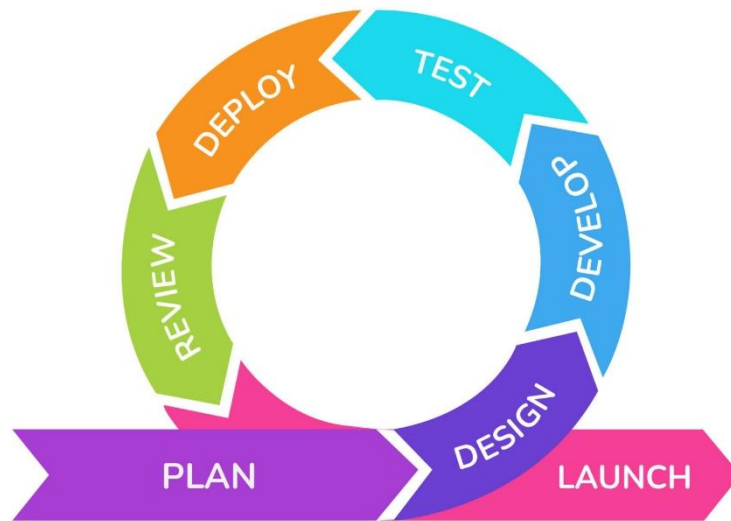
The SDLC provides a framework for the systematic planning, development, and maintenance of software projects. It typically consists of several stages: requirements analysis, design, implementation, testing, deployment, and maintenance. Each stage is interdependent, and the successful completion of one phase lays the groundwork for the next. The development phase focuses on the actual coding and integration of system components, while the testing phase is dedicated to identifying and rectifying defects or inconsistencies within the system. Various testing methodologies, including unit testing, integration testing, and user acceptance testing, are employed to ensure that the system functions as intended and adheres to quality standards.

The researchers chose the Agile Model that is frequently regarded as the most suitable approach for an online ordering management system. This model highlights the importance of adaptability, ongoing enhancements, and input from users in creating a

responsive and easy-to-use system. Agile enables a process of continuous development, enabling the release of updates and enhancements regularly in response to user feedback and evolving needs.

Figure 10

Software Development Life Cycle – Agile Model



Note: Image retrieved from <https://www.krasamo.com/wp-content/uploads/agile-01-scaled.jpeg>

Plan - Establish project goals, and user stories, and prioritize features based on customer needs and business objectives. The team identifies stakeholders, gathers requirements, and prioritizes tasks based on customer needs. This step ensures a clear understanding of project goals and helps align expectations across the team and stakeholders.

Design – This phase outlines the system's architecture and user interface, focusing on a seamless user experience. It includes planning intuitive navigation, layout, and essential features such as a simulated payment processor for GCash and Maya transactions. The

researchers work closely to ensure the system is both functional and user-friendly to represent the payment transaction processing feature.

Develop - Implement the planned features and functionalities. Coding is implemented according to design specifications, adhering to coding standards. This phase often involves pair programming and code reviews, promoting collaboration and knowledge sharing among team members. This ensures the code is not only functional but also maintainable and scalable.

Test - Ensure the developed features work as intended and meet quality standards. This includes unit tests, integration tests, and user acceptance testing to ensure that the system meets the defined requirements. Continuous testing allows for immediate feedback, helping to maintain high-quality standards throughout the development process.

Deploy – The tested features will be set up for live online access using hosting with a cloud service. Careful monitoring will be conducted during this phase to identify any immediate issues that may arise post-deployment, ensuring that the user experience remains seamless.

Review - Evaluate the outcomes and gather feedback for improvement. The team demonstrates completed features to stakeholders, gathers feedback, and discusses any lessons learned through the Survey questionnaires utilizing the 5-point Likert Scale format. This quantitative approach helps in refining the process and adjusting plans based on insights gained.

Launch – The updated system or new features will be released to the company if they desire to use them. If this is required, the team will communicate the updates, provide training sessions to help users navigate the system, and offer clear documentation and support. This will facilitate user adoption and satisfaction. A structured yet adaptable approach will enable

continuous improvement and alignment with user needs throughout the development process.

Each phase in the Agile model contributes to a collaborative, adaptive development process that ensures the online ordering management system meets user needs while allowing for continuous improvement, as shown in Figure 10.

Software Evaluation

In parallel, the Software Evaluation Plan outlines the criteria and metrics for assessing the system's performance based on the prescribed criteria. This plan defines the objectives of the evaluation process, specifying the types of tests to be conducted, the tools and techniques to be utilized, and the timeline for completion. It also includes user feedback mechanisms, which are vital for understanding how well the system meets user needs and expectations. By establishing clear evaluation criteria and systematically gathering feedback, the Software Evaluation Plan aims to facilitate continuous improvement of the application, ensuring it delivers value to its users.

Table 6

5-point Likert Scale Example Software Evaluation Results

No. of Respondents (R)	Criteria (C)	R*C
3	1	3
8	2	16
5	3	15
9	4	36
5	5	25
Total Respondents = 30		Total of R*C = 95

From Table 6, as a sample, there were 3 people who strongly disagreed, getting the product of R and C, $3*1 = 3$.

8 people who disagreed, getting the product of R and C, $8*2 = 16$.

5 people who somehow agreed, getting the product of R and C, $5*3 = 15$.

9 people who agreed, getting the product of R and C, $9*4 = 36$.

And there were 5 people who strongly agreed, getting the product of R and C, $5*5 = 25$.

Getting the total no. of respondents = 30.

Getting the total of $R*C = 95$.

To apply the formula in getting the results:

$$\text{Result} = \frac{\text{Total } R*C}{\text{Total } R} = \frac{95}{30} = 3.17 \quad \text{Somehow Agree}$$

Likert Scale Interpretation:

1 – Strongly Disagree

2 – Disagree

3 – Somehow Agree

4 – Agree

5 – Strongly Agree

Note that if the result falls on:

1-1.50 = Strongly Disagree

1.51-2.50 = Disagree

2.51-3.50 = Somehow Agree

3.51-4.5 = Agree

4.51-5.0 = Strongly Agree

The result shows that the respondents **somehow agree** based on the criteria set by the developer. However, the target is **3.51 or above**.

A total of 10 respondents participated in the Software Evaluation; these respondents consist of the stakeholders, such as (1) Owner, (3) Staff, (1) 3D Artist, and (5) invited friends to act as the Customers, who contribute to assessing the overall performance of the proposed system. A Software Evaluation Sheet was provided to allow scoring based on the 5-point Likert Scale and questions prepared per the Criteria. A sample Software Evaluation Form is attached in Appendix E, as well as some Actual Software Evaluation Forms in Appendix F.

Results and Discussion

This section presents the results of the evaluation conducted on the Web-based 3D Printing Service Ordering Management System. The system was assessed by ten respondents using a 5-point Likert scale with the following interpretation: *Strongly Agree* (4.51–5.00), *Agree* (3.51–4.50), *Somehow Agree* (2.51–3.50), *Disagree* (1.51–2.50), and *Strongly Disagree* (1.00–1.50). The evaluation focused on five major criteria, namely: Functionality, Reliability, Usability, Maintainability, and Portability.

The results summarize the respondents' perceptions of how well the system meets its intended objectives and the extent to which it addresses the operational needs of the company. Each table presents the mean scores and their corresponding interpretations per criterion, followed by an overall summary. These findings provide insights into the system's strengths and areas that may require further enhancement.

Table 7

Summary of Mean Scores of Functionality

Item	Mean Interpretation	
1. The system has a login facility to prevent unauthorized access.	4.4	Agree
2. The system generates correct information based on input data.	4.3	Agree
3. The system saves data in the database securely.	4.5	Agree
4. The system calculates profit accurately based on the input data.	4.3	Agree
5. The system generates accurate sales reports for analysis.	4.4	Agree
6. The system tracks order statuses and updates them in real-time.	4.5	Agree
7. The system accommodates inquiries and customer feedback efficiently.	4.3	Agree
8. The system is user-friendly and easy to navigate for both the administrator and clients.	4.4	Agree
9. The system allows users to submit 3D printing orders smoothly.	4.5	Agree
10. The system ensures backup and recovery of order and client data.	4.3	Agree
Overall Mean	4.40	Agree

The functionality criterion obtained an overall mean of **4.40**, interpreted as **Agree** as show in Table 7. This indicates that respondents strongly agreed that the system effectively provides essential functions such as secure login access, accurate information processing, real-time order tracking, and reliable backup and recovery mechanisms. The highest ratings were given to the system's ability to save data securely and track orders in real-time, suggesting that the system's core features are well-implemented and meet user needs.

Table 8

Summary of Mean Scores of Reliability

Item	Mean Interpretation	
1. The system is able to handle errors without crashing.	4.3	Agree
2. The system remains stable even with multiple users accessing it simultaneously.	4.2	Agree
3. The system ensures that data saved is accurate and consistent across sessions.	4.3	Agree
4. The system's error logs and recovery mechanisms are functional and easy to use.	4.2	Agree
5. The system can handle heavy loads of data and transactions efficiently.	4.3	Agree
Overall Mean	4.26	Agree

The reliability criterion garnered an overall mean of **4.26**, interpreted as **Agree** as shown in Table 8. This result implies that the respondents recognized the system as highly

stable and capable of managing multiple users and heavy transactions without errors or crashes. The findings also highlight that data accuracy and consistency are well-maintained across sessions, ensuring the dependability of the system during operations.

Table 9

Summary of Mean Scores of Usability

Item	Mean Interpretation	
1. The system can be easily learned by different users.	4.3	Agree
2. The system is easy to operate with minimal training.	4.2	Agree
3. The system has intuitive dropdown options for easy user selection.	4.2	Agree
4. The system responds quickly to user actions.	4.3	Agree
5. The system allows for easy data entry without requiring too many steps.	4.2	Agree
6. The system is accessible on different devices and screen sizes.	4.1	Agree
7. The system provides appropriate feedback after actions (e.g., saving data).	4.2	Agree
8. The system interface looks visually appealing and organized.	4.3	Agree
Overall Mean	4.23	Agree

The usability criterion achieved an overall mean of **4.23**, interpreted as **Agree** as shown in Table 9. Respondents agreed that the system is intuitive and user-friendly, with features that facilitate learning and minimize the need for extensive training. Users

particularly appreciated the system's visually appealing interface, organized layout, and quick responsiveness to actions, which contribute to a positive user experience.

Table 10

Summary of Mean Scores of Maintainability

Item	Mean Interpretation	
1. System errors related to client orders can be easily diagnosed.	3.9	Agree
2. The system can be easily tested for various client orders, payments, and inquiries.	4.0	Agree
3. The system has dropdown options for users to select 3D printing services and specifications (helpful for system maintenance).	3.9	Agree
4. The system can identify and log errors related to order processing, payment tracking, and client management.	4.0	Agree
5. The system's documentation and error logs are accessible for troubleshooting and maintenance by the admin.	4.0	Agree
Overall Mean	3.96	Agree

Maintainability received an overall mean of **3.96**, interpreted as **Agree** as shown in Table 10. This suggests that the system is easy to maintain, diagnose, and troubleshoot. Respondents noted that the availability of system documentation and error logs assists administrators in addressing technical issues. Although the ratings were slightly lower compared to other criteria, the results still indicate that the system supports efficient maintenance processes.

Table 11*Summary of Mean Scores of Portability*

Item	Mean Interpretation	
1. The system functions the same way across different web browsers.	4.1	Agree
2. The system is easily accessible on various devices (laptops, desktops, mobile devices).	4.0	Agree
3. The system complies with portability standards, ensuring smooth transitions across different environments.	4.1	Agree
4. The system's interface adjusts responsively to different screen sizes.	4.0	Agree
5. The system supports multiple languages and can be localized easily.	4.1	Agree
6. The system's performance is consistent regardless of internet speed variations.	4.0	Agree
7. The system can easily replace other existing software without complications.	4.1	Agree
8. The system allows users to move data or settings between platforms effortlessly.	4.0	Agree
9. The system can be used on different operating systems.	4.1	Agree
Overall Mean	4.09	Agree

The portability criterion attained an overall mean of **4.09**, interpreted as **Agree** as shown in Table 11. This demonstrates that respondents agreed the system performs

consistently across different devices, browsers, and operating systems. Users also recognized that the system adapts well to varying screen sizes and internet conditions, ensuring accessibility in various environments.

Table 12

Summary of Mean Scores per Criteria and Overall Evaluation

Criteria	Number of Items	Mean Interpretation	
Functionality	10	4.40	Agree
Reliability	5	4.26	Agree
Usability	8	4.23	Agree
Maintainability	5	3.96	Agree
Portability	9	4.09	Agree
Overall Mean	37	4.21	Agree

Table 12 presents the consolidated mean scores of all five criteria used to evaluate the Web-based 3D Printing Service Ordering Management System. Among the criteria, Functionality obtained the highest mean score of 4.40 (Agree), indicating that respondents strongly agreed the system effectively delivers its intended features, such as secure login, accurate data processing, and real-time order tracking. This was closely followed by Reliability (4.26) and Usability (4.23), both interpreted as Agree, suggesting that the system is stable even with multiple users, provides consistent performance, and is intuitive and easy to use.

On the other hand, Portability received a mean score of 4.09 (Agree), showing that the system generally works well across different devices, browsers, and operating systems. Maintainability obtained the lowest mean score of 3.96 (Agree), indicating that while the system supports troubleshooting and maintenance processes, this area may still have room for improvement compared to the other criteria.

The overall mean score across all criteria was 4.21 (Agree). This result reflects that respondents were highly satisfied with the system's overall performance. The findings confirm that the system effectively meets the desired standards of quality and efficiency, addressing previous operational challenges such as slow response times, inconsistent pricing, and poor record-keeping.

CHAPTER IV

CONCLUSION AND RECOMMENDATION

In summation, this study successfully developed and evaluated a Web-based 3D Printing Service Ordering Management System tailored for TechWorkz, effectively addressing the identified challenges of inefficient communication, inconsistent pricing, and disorganized record-keeping. The system was designed to streamline order processing, enhance customer interaction, and improve overall operational efficiency. Through comprehensive requirement analysis, cost-benefit analysis, and data gathering procedures, the necessity and feasibility of the proposed system were validated. The implementation of a centralized online platform, featuring modules for user management, order tracking, payment monitoring, and reporting, successfully met the objectives outlined in Chapter I. Specifically, the system facilitated a seamless transition from Facebook Messenger-based inquiries to a dedicated web platform, significantly improving accessibility and communication.

Furthermore, it provided a centralized platform for staff to efficiently manage orders and track project progress, thereby enhancing workflow efficiency. The automation of order processing and cost estimation reduced manual intervention, minimized errors, and ensured consistent pricing. The system's ability to generate comprehensive reports enabled informed decision-making for the business owner, providing valuable insights into sales and order management. Evaluation using the 5-point Likert Scale confirmed the system's functionality, usability, and positive user feedback, demonstrating its effectiveness in meeting the needs of both clients and staff. The cost-benefit analysis revealed significant long-term financial benefits, with projected savings over five years justifying the initial investment. The system's

robust architecture, visualized through various diagrams, ensured a scalable and reliable platform. The significance of this study extends beyond TechWorkz, providing a model for efficient online order management that demonstrates how technology can drive business success in the competitive 3D printing industry. It empowers small-to-medium enterprises to adopt advanced solutions, enhancing customer satisfaction and operational efficiency.

Recommendations

Based on the study's findings, several recommendations are made to further enhance the system and guide future research. To further streamline the ordering process, future development should focus on integrating direct payment processing capabilities within the system, eliminating the need for external payment methods and reducing processing times. The system should also be expanded to include physical logistics management features, enabling clients to track shipping and delivery and providing a more comprehensive order tracking experience.

Integration with existing inventory management systems would improve overall business efficiency by enabling real-time tracking of materials and supplies. Developing a mobile application version of the system would enhance accessibility and user convenience, allowing clients to manage orders and track progress on their mobile devices. Further research should focus on assessing the system's scalability and performance under high-volume conditions to ensure it can handle increased demand without compromising efficiency.

Additionally, future iterations should include more variable inputs to the 3D printing calculator module, such as material density, infill percentage, support structure requirements, and post-processing costs. This enhanced calculator would provide clients with more

accurate and detailed cost estimations, improving transparency and customer satisfaction. Future research could explore the application of emerging technologies, such as AI and machine learning, to enhance the system's capabilities, particularly in areas like automated design analysis and predictive maintenance. For instance, AI algorithms could be developed to analyze uploaded 3D models for potential printing issues, suggest optimizations, and provide real-time feedback to clients. Machine learning could also be used to predict printer maintenance needs, minimizing downtime and ensuring consistent output quality.

Furthermore, studies on the impact of web-based systems on customer loyalty and retention in the 3D printing industry would provide valuable insights, contributing to the broader understanding of technology's role in the industry's growth. This could include research on user experience design, personalized service offerings, and the integration of customer feedback mechanisms to foster long-term client relationships. Additionally, examining the feasibility of integrating augmented reality (AR) or virtual reality (VR) technologies for design visualization and product previews could offer innovative ways to enhance the customer experience and differentiate TechWorkz in the market.

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APPENDICES
APPENDIX A
Transcript of Interview

Background: The owner of Techworkz 3D printing services would like to shift his clients from different social media platform ordering to a web-based centralized ordering system that can track payments, accommodate inquiries and to improve and enhance the ordering process.

In this part of the case, Alfred Magno (business owner), Samuel Chisom Ogu and Abdulaziz Noor Fitwi (AMA OED Students) are talking about different problems encountered from the existing business process and discuss the proposed new system and the project's feasibility.

Participants: Alfred Magno, Samuel Chisom Ogu, and Abdulaziz Noor Fitwi

Location: Online Meeting via Google Meet, Monday afternoon, October 30th, 2024

Project Status: Samuel and Abdulaziz presented their proposed system to Mr. Alfred Magno after the discovery of the problems from the existing system and discussed the features of the proposed system that can solve those problems.

Samuel: Hi, Mr. Magno. Are you ready to get started?

Mr. Magno: Yes

Samuel: Mr. Magno. Where are you located?

Mr. Magno: *I am located here in Cavite.*

Samuel: What type of business do you have?

Mr. Magno: *3D printing services, we also do paintings of the models, and 3d modeling commissions.*

Samuel: We are students of AMA University Online Education taking up Bachelor of Science in Information Technology, we are now on a Capstone project which we are required to find company that we can propose a new system to improve their business process, we were referred to you by a friend who knows your services and we would like to ask if we can ask your permission to proposed our project with your company?

Mr. Magno: *Yes, It's ok.*

Samuel: Ok Sir, Thank you for accepting us to interview you right now.

Samuel: To begin with, can you tell us how you conduct your day-to-day business transactions in accommodating your client's request for 3D printing?

Mr. Magno: *We usually get a lot of inquiries from different community and groups from Facebook, they would send us a message inquiry asking and requesting for a specific 3d printing service or project, our staff usually would reply to their inquiries providing them information about their inquiries then once they agreed to the terms and conditions, we then finalize the order by either asking them their advance payment or down payment before we proceed to print the 3d model they'd like to have.*

Abdulaziz: What are the problems you encounter with your current business transactions?

Mr. Magno: *Oh yes there's a lot, especially we received numerous messages from Facebook messenger and not all of them can be accommodated promptly, some inquiries are only consuming time because they have a lot of questions regarding our services and end up those clients do not proceed to order or get our services, also, we encountered problems with payments, some clients get away without payment and the 3d model has already been printed and they're gone?*

Abdulaziz: How many employees do you have right now?

Mr. Magno: *I only have 2 staff right now who answer inquiries and orders online. Sometimes I have 2 personnel that operate 3D printers and a 3d artist.*

Abdulaziz: How many 3D printer machines do you have right now?

Mr. Magno: *I only have 5 3D printers.*

Abdulaziz: How many 3D printer machines do you have right now?

Mr. Magno: *I only have 5 3D printers.*

Samuel: Are you open to have a centralized system where your clients can go to and all of their Frequently Asked Questions are there so you don't need to answer every inquiry?

Mr. Magno: *Yes of course, and also if there can be a system that can shift them from sending inquiries in FB Messenger to a centralized ordering system that would be great!*

Abdulaziz: What other features do you think this new system should have?

Mr. Magno: *I'd also like to track payments and sales because sometimes I am not in the shop only the staffs would report to me, I also would like to include where we can showcase our portfolio or gallery of 3d printed models so clients can view it, I'd also like to have something like the clients can choose ready to print 3d models and chat with our staffs.*

Samuel: Would you accept us to propose to you a new system which we will be discussing to you in our next meeting once approved by the school?

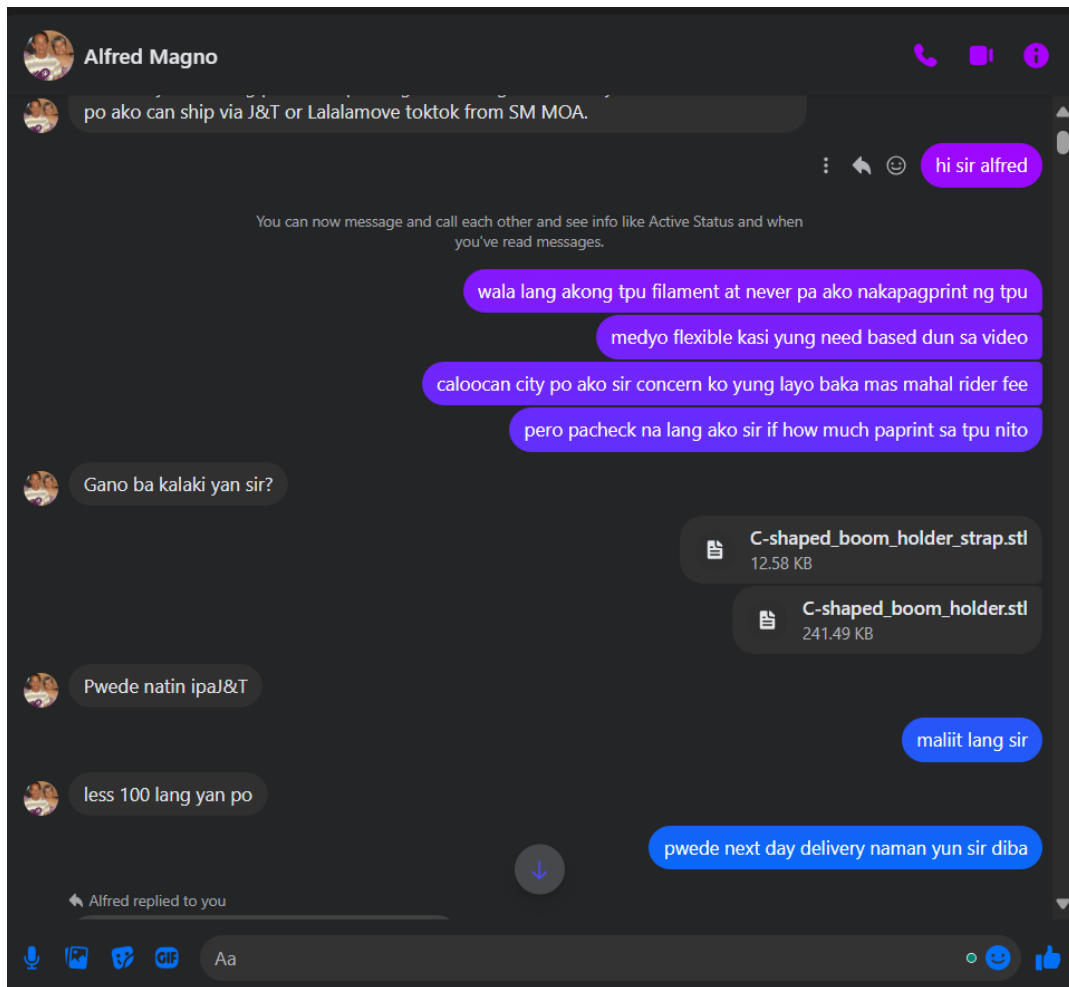
Mr. Magno: *Yes, I am excited to know your proposal!*

Samuel: We also appreciate your time and kindness Mr. Magno!

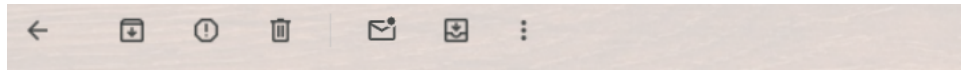
Abdulaziz: Thank you very much, Sir!

APPENDIX B

Manual Forms Used by the Company



Client Ordering via Messenger



Fwd: 3D scanned head Inbox x



Jerome Magno <jarmsanity@gmail.com>

to me ▾

----- Forwarded message -----

From: **Jerome Magno** <jarmsanity@gmail.com>

Date: Thu, 20 Jul 2023, 5:36 pm

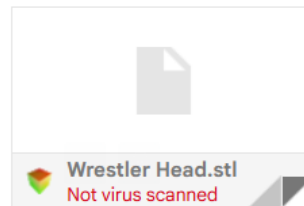
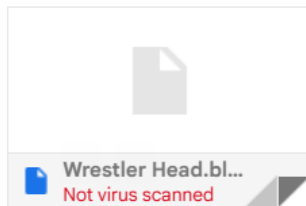
Subject: 3D scanned head

To: <pjtugade@yahoo.com>

 **Wrestler Head.blend**

 **Wrestler Head.stl**

2 Attachments • Scanned by Gmail ⓘ  Add all to Drive



Client Transaction via Email

	A	B	C	D	E	F	G	H	I
1	TECHWORKZ 3D PRINTS								
2	ORDERS AND COMMISSIONS								
3									
4									
5	DATE	DESCRIPTION	CLIENTS NAME	CONTRACT PRICE	DOWN PAYMENT	UNPAID	BALANCE PAID	TOTAL SALES	
6	23/05/2024	EHONDA STREET FIGHTER 8"	Warren Valenzuela	P 3,800.00	P 1,900.00	P 1,900.00		P 1,900.00	
7	29/05/2024	SEAHORSE	Jamie Trinidad	P 1,250.00	P 750.00	P -	P 500.00	P 1,250.00	
8	29/05/2024	PEGASUS FOR VALKYRIE	John Paul Borromeo	P 1,300.00		P 1,300.00		P -	
9	29/05/2024	KOBE BRYANT STATUE 12"	Bro. Dek Bulseco	P 4,500.00	P 2,500.00	P 2,000.00		P 2,500.00	
10						P -		P -	
11						P -		P -	
12						P -		P -	
13						P -		P -	
14						P -		P -	
15						P -		P -	
16						P -		P -	
17						P -		P -	

Excel Sheet 3D Prints log

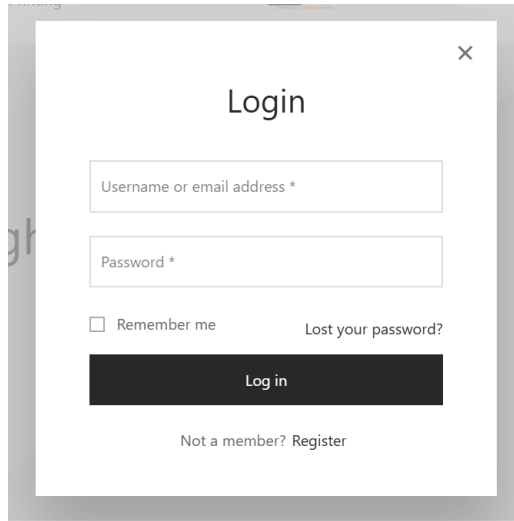
APPENDIX C

PHOTOS INSIDE THE 3D PRINTING AREA



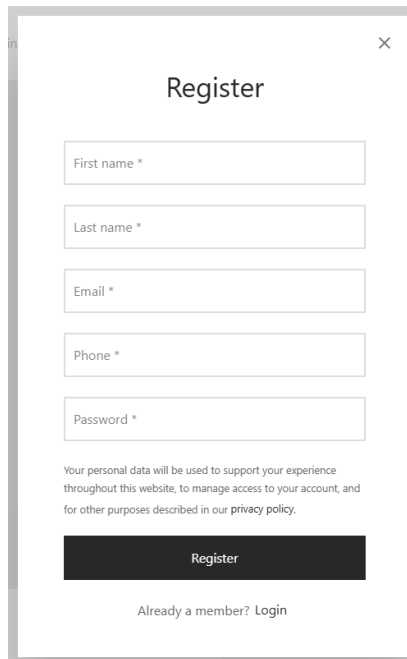
APPENDIX D

Forms and Reports of the System



A screenshot of a login form within a modal window. The modal has a close button (X) in the top right corner. The title "Login" is centered at the top. Below the title are two input fields: "Username or email address *" and "Password *". Below these fields is a checkbox labeled "Remember me" and a link "Lost your password?". A dark button labeled "Log in" is centered below the checkbox. At the bottom, there is a link "Not a member? Register".

Login Form



A screenshot of a registration form within a modal window. The modal has a close button (X) in the top right corner. The title "Register" is centered at the top. Below the title are five input fields: "First name *", "Last name *", "Email *", "Phone *", and "Password *". Below these fields is a paragraph of text: "Your personal data will be used to support your experience throughout this website, to manage access to your account, and for other purposes described in our [privacy policy](#)." A dark button labeled "Register" is centered below the text. At the bottom, there is a link "Already a member? Login".

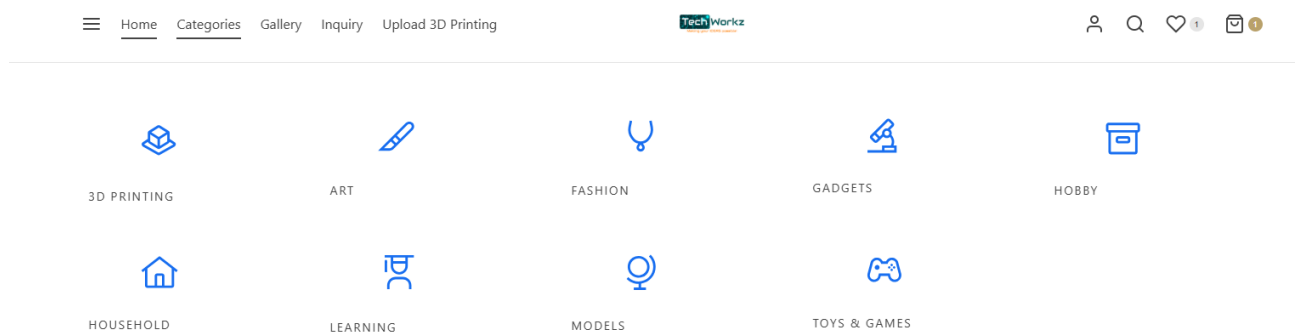
Registration Form

Lost password

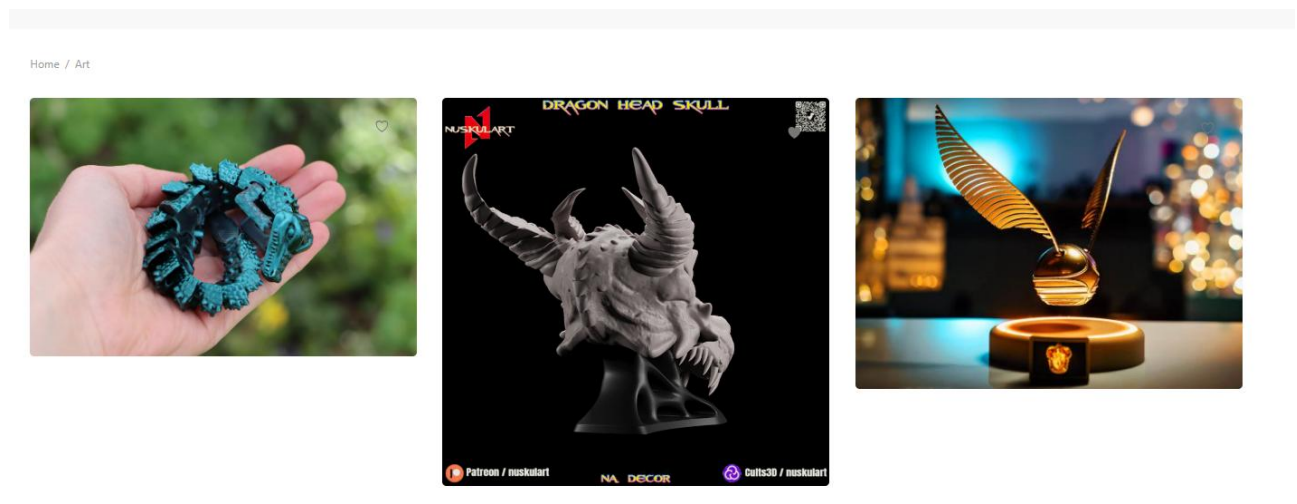
Lost your password? Please enter your username or email address. You will receive a link to create a new password via email.

Reset password

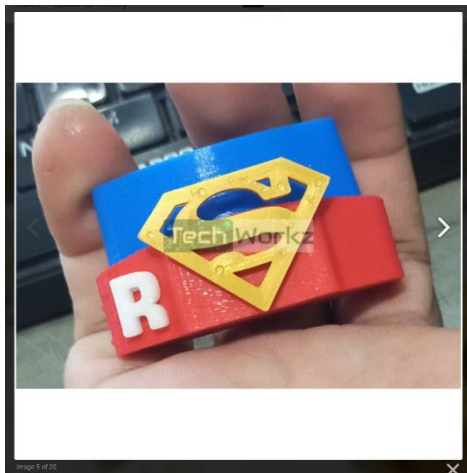
Lost password form



Catalog by Category



Catalog Sample



Sample Works in the Gallery

Inquiry

Inquiry

Fields marked with an * are required

First Name *

Last Name *

Email *

Phone *

Message *

Send

Guest Inquiry Form

Upload 3D Printing

Upload 3D Printing

Name*

Email*

Phone*

Upload 3D STL*
Choose File No file chosen

Message*

Send

Upload 3D Printing file Form

Cart

PRODUCT	TOTAL
 Articulated Crocodile 1,100.00 ₱ Material type: PLA(Poly(lactic Acid)) - 1 + Remove Item	1,100.00 ₱

CART TOTALS

[Add a coupon](#)

Subtotal 1,100.00 ₱

Delivery 300.00 ₱
Fast Delivery

Delivers to **METRO MANILA, PHILIPPINES**

☒ Fast Delivery 300.00 ₱
☐ Regular Delivery 200.00 ₱

Total 1,400.00 ₱

[Proceed to Checkout](#)

Cart Page



Have a coupon? [Click here to enter your code](#)

Billing details

[Back to Cart](#)

First name *	Last name *
Company name (optional)	
Country / Region * Philippines	
Street address *	
Apartment, suite, unit, etc.	
Town / City *	
State / County * Metro Manila	
Postcode / ZIP *	
Phone *	
Email address *	

Billing details



Articulated Crocodile - PLA(Polylactic Acid)
x 1
1.100,00 ₱

SUBTOTAL

1.100,00 ₱

SHIPPING

☒ Fast Delivery: 300,00 ₱
☐ Regular Delivery: 200,00 ₱

TOTAL

1.400,00 ₱

☒ GCash via PayMongo


Simple and easy payments via GCash.

☐ Maya via PayMongo


Your personal data will be used to process your order, support your experience throughout this website, and for other purposes described in our [privacy policy](#).

☐ I have read and agree to the website terms and conditions *

Place order

Payment Options



GCash Test Payment Page

This is a test payment of ₱ 1,400.00.

Authorize Test Payment

Fail / Expire Test Payment

Gcash Test Payment via Paymongo

Order received

Thank You! Order has been received.

Order number:	Date:	Email:	Total:	Payment method:
4201	August 6, 2025	emoran@oed.com.ph	1,400,00 ₱	GCash via PayMongo

Order details

Articulated Crocodile - PLA(Polylactic Acid) 	1,100,00 ₱
Material type: PLA(Polylactic Acid)	
SUBTOTAL:	1,100,00 ₱
SHIPPING:	300,00 ₱ via Fast Delivery
PAYMENT METHOD:	GCash via PayMongo
TOTAL:	1,400,00 ₱

Billing address

Bruce Wayne
Wayne Mansion
Gotham City
Metro Manila
1234
0999998888
emoran@oed.com.ph

Shipping address

Bruce Wayne
Wayne Mansion
Gotham City
Metro Manila
1234

Order Placed and Received

APPENDIX E

SOFTWARE EVALUATION FORM SAMPLE

Research Title	Web-Based 3D Printing Service ordering Management System
Name of Researcher	Samuel Chisom Ogu / Abdulaziz Noor Fitwi
Name of Evaluator	
Email Address	
Designation/Position in the Company	
Company Name	
Company Address	

Instructions:

Kindly evaluate the Web-Based 3D Printing Service Ordering Management System based on the numerical rating scale below. Please check the box that corresponds to your rating for each criterion.

Numerical Rating	Description
1	Strongly Disagree
2	Disagree
3	Somehow Agree
4	Agree
5	Strongly Agree

Functionality	5	4	3	2	1
1. The system has a login facility to prevent unauthorized access.					
2. The system generates correct information based on input data.					
3. The system saves data in the database securely.					
4. The system calculates profit accurately based on the input data.					
5. The system generates accurate sales reports for analysis					
6. The system tracks order statuses and updates them in real-time.					
7. The system accommodates inquiries and customer feedback efficiently.					
8. The system is user-friendly and easy to navigate for both the administrator and clients.					
9. The system allows users to submit 3D printing orders smoothly.					
10. The system ensures backup and recovery of order and client data.					

Reliability	5	4	3	2	1
1. The system is able to handle errors without crashing.					
2. The system remains stable even with multiple users accessing it simultaneously.					
3. The system ensures that data saved is accurate and consistent across sessions.					
4. The system's error logs and recovery mechanisms are functional and easy to use.					
5. The system can handle heavy loads of data and transactions efficiently.					

Usability	5	4	3	2	1
1. The system can be easily learned by different users.					
2. The system is easy to operate with minimal training.					
3. The system has intuitive dropdown options for easy user selection.					
4. The system responds quickly to user actions.					
5. The system allows for easy data entry without requiring too many steps					
6. The system is accessible on different devices and screen sizes.					
7. The system provides appropriate feedback after actions (e.g., saving data)					
8. The system interface looks visually appealing and organized.					

Maintainability	5	4	3	2	1
1. System errors related to client orders can be easily diagnosed.					
2. The system can be easily tested for various client orders, payments, and inquiries.					
3. The system has dropdown options for users to select 3D printing services and specifications (helpful for system maintenance).					
4. The system can identify and log errors related to order processing, payment tracking, and client management.					
5. The system's documentation and error logs are accessible for troubleshooting and maintenance by the admin.					

Portability	5	4	3	2	1
1. The system functions the same way across different web browsers (Chrome, Brave, Safari, etc.).					
2. The system is easily accessible on various devices (laptops, desktops, mobile devices).					
3. The system complies with portability standards, ensuring smooth transitions across different environments.					
4. The system's interface adjusts responsively to different screen sizes (desktop, tablet, mobile).					
5. The system supports multiple languages and can be localized easily for different regions.					
6. The system's performance is consistent regardless of internet speed variations.					
7. The system can easily replace other existing software without complications					
8. The system allows users to move data or settings between platforms effortlessly.					
9. The system can be used on different operating systems (Windows, Mac, etc.).					

APPENDIX F

ACTUAL SOFTWARE EVALUATION FORM

SOFTWARE EVALUATION FORM

Research Title	Web-Based 3D Printing Service Ordering Management System
Name of Researcher	Samuel Chisom Ogu / Abdulaziz Noor Fitwi
Name of Evaluator	Rechelle Nueva
Email Address	mevarechelle@gmail.com
Designation/Position in the Company	Hr officer / Sales Staff
Company Name	Techworkz 3D Printing Service
Company Address	Sunngcrest Village, Salita II, Dasmarinas City, Cavite

instructions:

Kindly evaluate the Web-Based 3D Printing Service Ordering Management System based on the numerical rating scale below. Please check the box that corresponds to your rating for each criterion.

Numerical Rating	Description
1	Strongly Disagree
2	Disagree
3	Somehow Agree
4	Agree
5	Strongly Agree

Functionality	5	4	3	2	1
1. The system has a login facility to prevent unauthorized access.	✓				
2. The system generates correct information based on input data.	✓				
3. The system saves data in the database securely.	✓				
4. The system calculates profit accurately based on the input data.	✓				
5. The system generates accurate sales reports for analysis	✓				
6. The system tracks order statuses and updates them in real-time.		✓			
7. The system accommodates inquiries and customer feedback efficiently.		✓			
8. The system is user-friendly and easy to navigate for both the administrator and clients.		✓			
9. The system allows users to submit 3D printing orders smoothly.		✓			
10. The system ensures backup and recovery of order and client data.			✓		

Reliability	5	4	3	2	1
1. The system is able to handle errors without crashing.		✓			
2. The system remains stable even with multiple users accessing it simultaneously.	✓				
3. The system ensures that data saved is accurate and consistent across sessions.	✓				
4. The system's error logs and recovery mechanisms are functional and easy to use.			✓		
5. The system can handle heavy loads of data and transactions efficiently.		✓			

Usability	5	4	3	2	1
1. The system can be easily learned by different users.	✓				
2. The system is easy to operate with minimal training.		✓			
3. The system has intuitive dropdown options for easy user selection.	✓				
4. The system responds quickly to user actions.		✓			
5. The system allows for easy data entry without requiring too many steps	✓				
6. The system is accessible on different devices and screen sizes.		✓			
7. The system provides appropriate feedback after actions (e.g., saving data)		✓			
8. The system interface looks visually appealing and organized.		✓			

Maintainability	5	4	3	2	1
1. System errors related to client orders can be easily diagnosed.		✓			
2. The system can be easily tested for various client orders, payments, and inquiries.		✓			
3. The system has dropdown options for users to select 3D printing services and specifications (helpful for system maintenance).	✓				
4. The system can identify and log errors related to order processing, payment tracking, and client management.			✓		
5. The system's documentation and error logs are accessible for troubleshooting and maintenance by the admin.			✓		

Portability	5	4	3	2	1
1. The system functions the same way across different web browsers (Chrome, Brave, Safari, etc.).		✓			
2. The system is easily accessible on various devices (laptops, desktops, mobile devices).	✓				
3. The system complies with portability standards, ensuring smooth transitions across different environments.		✓			
4. The system's interface adjusts responsively to different screen sizes (desktop, tablet, mobile).		✓			
5. The system supports multiple languages and can be localized easily for different regions.		✓			
6. The system's performance is consistent regardless of internet speed variations.			✓		
7. The system can easily replace other existing software without complications		✓			
8. The system allows users to move data or settings between platforms effortlessly.		✓			
9. The system can be used on different operating systems (Windows, Mac, etc.).	✓				

SOFTWARE EVALUATION FORM

Research Title	Web-Based 3D Printing Service Ordering Management System
Name of Researcher	Samuel Chisom Ogu / Abdulaziz Noor Fitwi
Name of Evaluator	ALFRED MAEND
Email Address	jarmcanity@gmail.com
Designation/Position in the Company	OWNER
Company Name	TECHWORKZ 3D PRINTING SERVICE
Company Address	SUNNYCREST VILLAGE, SALITA II, PASMARINAS CITY, CAVITE

Instructions:

Kindly evaluate the Web-Based 3D Printing Service Ordering Management System based on the numerical rating scale below. Please check the box that corresponds to your rating for each criterion.

Numerical Rating	Description
1	Strongly Disagree
2	Disagree
3	Somehow Agree
4	Agree
5	Strongly Agree

Functionality	5	4	3	2	1
1. The system has a login facility to prevent unauthorized access.	✓				
2. The system generates correct information based on input data.	✓				
3. The system saves data in the database securely.	✓				
4. The system calculates profit accurately based on the input data.	✓				
5. The system generates accurate sales reports for analysis	✓				
6. The system tracks order statuses and updates them in real-time.	✓				
7. The system accommodates inquiries and customer feedback efficiently.	✓				
8. The system is user-friendly and easy to navigate for both the administrator and clients.	✓				
9. The system allows users to submit 3D printing orders smoothly.		✓			
10. The system ensures backup and recovery of order and client data.			✓		

Reliability	5	4	3	2	1
1. The system is able to handle errors without crashing.		✓			
2. The system remains stable even with multiple users accessing it simultaneously.		✓			
3. The system ensures that data saved is accurate and consistent across sessions.	✓				
4. The system's error logs and recovery mechanisms are functional and easy to use.		✓			
5. The system can handle heavy loads of data and transactions efficiently.		✓			

Usability	5	4	3	2	1
1. The system can be easily learned by different users.		✓			
2. The system is easy to operate with minimal training.		✓			
3. The system has intuitive dropdown options for easy user selection.	✓				
4. The system responds quickly to user actions.	✓				
5. The system allows for easy data entry without requiring too many steps	✓				
6. The system is accessible on different devices and screen sizes.		✓			
7. The system provides appropriate feedback after actions (e.g., saving data)		✓			
8. The system interface looks visually appealing and organized.		✓			

Maintainability	5	4	3	2	1
1. System errors related to client orders can be easily diagnosed.		✓			
2. The system can be easily tested for various client orders, payments, and inquiries.	✓				
3. The system has dropdown options for users to select 3D printing services and specifications (helpful for system maintenance).	✓				
4. The system can identify and log errors related to order processing, payment tracking, and client management.		✓			
5. The system's documentation and error logs are accessible for troubleshooting and maintenance by the admin.			✓		

Portability	5	4	3	2	1
1. The system functions the same way across different web browsers (Chrome, Brave, Safari, etc.).	✓				
2. The system is easily accessible on various devices (laptops, desktops, mobile devices).	✓				
3. The system complies with portability standards, ensuring smooth transitions across different environments.	✓				
4. The system's interface adjusts responsively to different screen sizes (desktop, tablet, mobile).		✓			
5. The system supports multiple languages and can be localized easily for different regions.		✓			
6. The system's performance is consistent regardless of internet speed variations.		✓			
7. The system can easily replace other existing software without complications		✓			
8. The system allows users to move data or settings between platforms effortlessly.		✓			
9. The system can be used on different operating systems (Windows, Mac, etc.).	✓				

APPENDIX G

Plagiarism Report



Plagiarism Checker X - Report

Originality Assessment

6%



Overall Similarity

Date: Aug 7, 2025 (07:21 pm)
Matches: 1267 / 19643 words
Sources: 40

Remarks: Low similarity detected, consider making necessary changes if needed.

Verify Report:
Scan this QR Code



APPENDIX H

User's Manual



User's Manual

*Web-Based 3D Printing Service Ordering Management System
For TechWorkz*

Online Education Department

October, 2025

**Abdulaziz N. Fitwi
Samuel C. Ogu**

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1.0 GENERAL INFORMATION

1.0 GENERAL INFORMATION

1.1 System Overview

*The **Web-Based 3D Printing Service Ordering Management System** is an online platform designed to streamline customer order processing, enhance communication, and centralize record-keeping for 3D printing services. The system enables clients to:*

- *Register for an account or browse as a guest.*
- *View ready-to-print 3D model catalogs and previous projects.*
- *Upload custom 3D model files for printing.*
- *Automatically calculate order costs using a built-in 3D Printing Calculator.*
- *Track the real-time status of orders.*
- *Communicate with staff via an integrated chat module.*
- *Receive automated email notifications about order updates and payments.*

System Architecture:

The system is web-based and hosted on a cloud server, accessible through any standard web browser. It follows a client-server architecture, where users interact with a graphical user interface (GUI) while the backend processes handle order management, file storage, and reporting.

User Access Mode:

- *Clients/Guests: Web browser access (desktop/laptop recommended).*
- *Staff/Admin: Secure login with role-based access controls.*

Responsible Organization:

TechWorkz 3D Printing Services – Sunnycrest Village, Salitran II, Dasmariñas City, Cavite, Philippines.

System Environment & Special Conditions:

- *Requires a stable internet connection (minimum 10 Mbps or above).*
- *Best viewed on modern browsers (Chrome, Edge, Firefox, Safari).*
- *Designed for desktop use, but accessible on mobile devices.*

1.2 Project References

- Ogu, S.C., & Fitwi, A.N. (2025). *Web-Based 3D Printing Service Ordering Management System for TechWork bz – Capstone Project, AMA University Online Education.*

1.3 Authorized Use Permission

This system is intended solely for the use of **TechWorkz 3D Printing Services**, its staff, and registered clients. Unauthorized access, duplication, or distribution of the system, its source code, data, or reports is strictly prohibited. Any attempt to modify or exploit the system without proper authorization will be subject to legal action.

1.4 Points of Contact

Name: Samuel C. Ogu Cellphone

Name: Abdulaziz N. Fitwi

No.: 09495214834 Email:

Cellphone No.: 09542265968

samuelogu01@gmail.com

1.5 Acronyms and Abbreviations

Acronym Meaning

3D - Three-Dimensional

AM - Additive Manufacturing

CAD - Computer-Aided Design

DBMS - Database Management System

GUI - Graphical User Interface

PLA - Polylactic Acid (3D printing filament)

ABS - Acrylonitrile Butadiene Styrene (3D printing filament)

2.0 SYSTEM SUMMARY

2.0 SYSTEM SUMMARY

The **Web-Based 3D Printing Service Ordering Management System** is a cloud-hosted platform designed to streamline 3D printing order management, automate pricing, track order progress, and facilitate direct communication between customers and staff. It replaces manual processes and Facebook Messenger interactions, allowing users to upload custom designs, choose ready-to-print models, receive cost estimates, and monitor orders in real time. The system features role-based access, guest browsing, integrated chat, email notifications, and reporting tools. Evaluated by 10 respondents including the owner, staff, a 3D artist, and invited customers, using a 5-point Likert scale, the system received high ratings across all criteria: Functionality 4.40, Reliability 4.26, Usability 4.23, Portability 4.09, and Maintainability 3.96, with an overall mean of 4.21 (Agree). These results confirm that the system effectively addresses operational challenges, providing a user-friendly, reliable, and accessible platform.

2.1 Hardware and Software Specifications

Client Devices - The system is accessible on desktop PCs, laptops, or mobile devices with a web browser and a stable internet connection. Some features may have limited display or functionality on smaller devices.

Cloud Hosting - The system is deployed on smarterasp.net to ensure online accessibility for all users.

Development Environment - The system was developed using a multi-core processor, 16 GB RAM, 250 GB SSD, Windows OS, Apache Web Server, MySQL, WordPress CMS, and programming languages including PHP, JavaScript, HTML, and CSS.

Payment Integration - Secure payment processing is enabled through PayMongo, GCash, and Maya Gateway.

2.2 User Access Levels

Admin (Owner/Administrator) - The admin has full control over the system. They can create and manage staff accounts, approve orders, verify payments, generate business analytics, and configure system settings.

Staff - Staff members handle customer orders, communicate with clients, upload gallery images, verify payments, and update the status of orders.

Registered Client - Registered clients can create an account, place 3D printing orders, upload custom designs, select from ready-made models, use the cost calculator, track order status, send and receive messages, and view transaction history.

Guest - Guests can browse the gallery, view ready-to-print models, read FAQs, and send inquiries through the contact form. They cannot place orders without registering.

2.3 Contingencies and Alternate Modes of Operation

In the event of system downtime due to website maintenance or server issues, clients will not be able to place orders or make payments until the system is restored. General internet issues on the company's side may affect internal operations, but as long as the website is online, clients can still place orders and complete payments. Staff can record and track orders manually if needed, and once the system is back online, they can update and process these orders. The system also performs daily automatic backups through the cloud hosting provider, keeping copies for 30 days to ensure data can be safely restored in case of loss or corruption.

3.0 GETTING STARTED

3.0 GETTING STARTED

This section provides step-by-step guidance on how to access and navigate the system from login to exit. It is written in clear, non-technical language for ease of use.

3.1 Logging On

1. *Open a Web Browser*

Recommended browsers: Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.

2. *Access the Login Page*

Click the Login button at the top right corner of the homepage.

3. *Enter Credentials*

- *Registered Clients/Staff/Admin: Enter your username and password.*
- *First-Time Users: Click Register to create a new account.*

4. *Click “Login”*

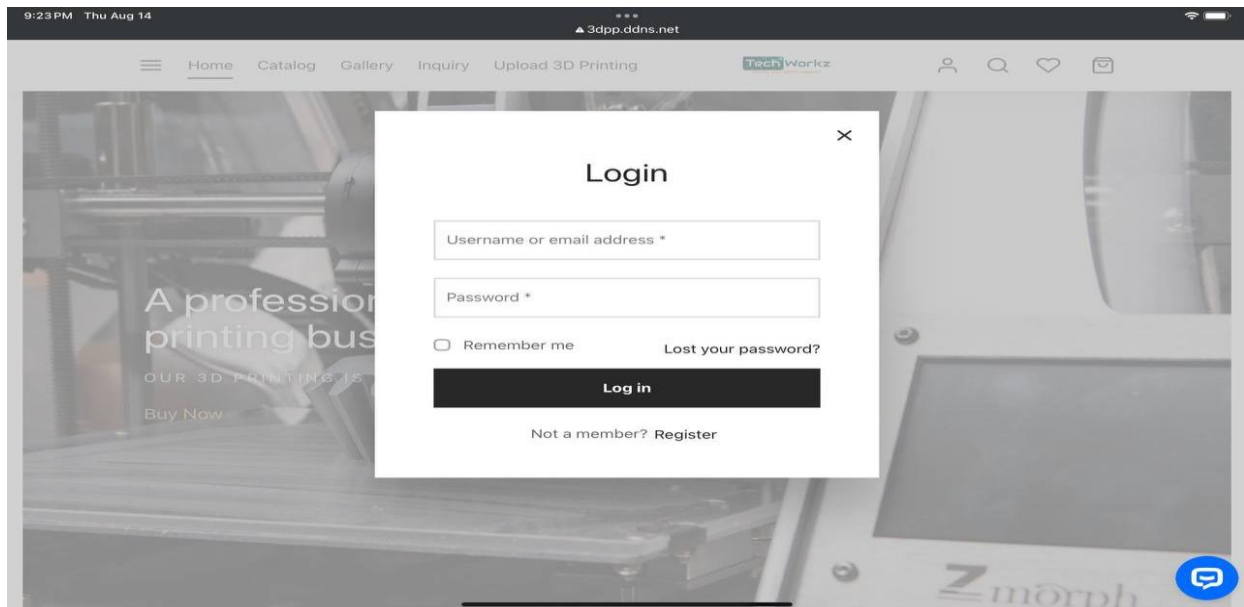
If the credentials are correct, you will be redirected to your dashboard.

5. *Guest Access*

Guests can browse galleries, FAQs, and ready-to-print models without logging in.

Note: Three consecutive failed login attempts will temporarily lock the account for security purposes.

Access the Login Page - Clients/Staff/Admin: Enter your username and password.



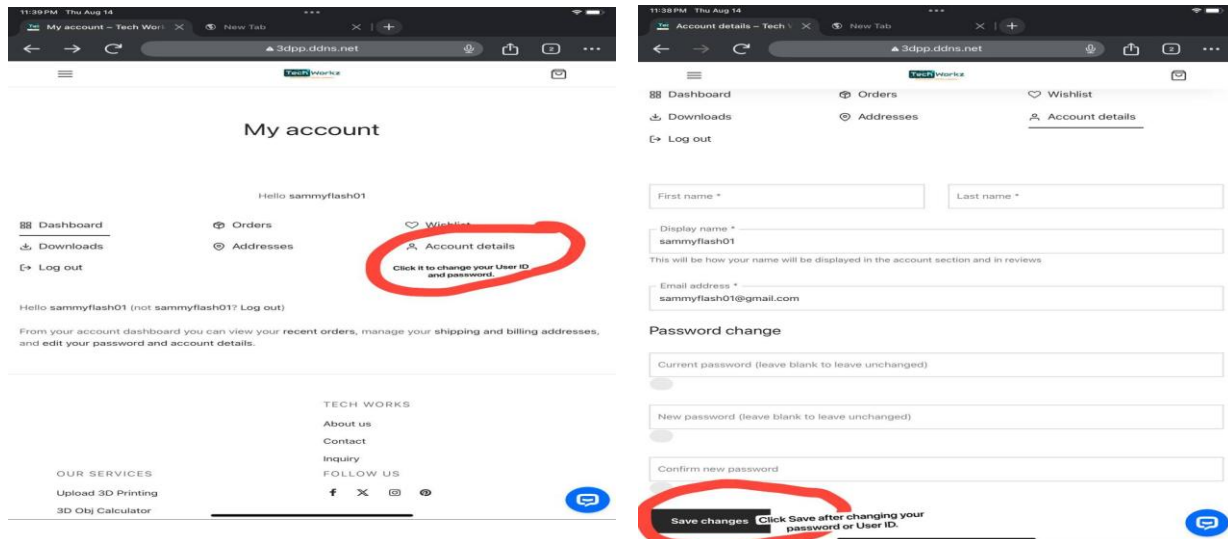
3.2 System Menu

The System Menu provides access to all the main features of the system, organized according to your user role. Registered clients can browse the gallery and ready-to-print models, upload custom orders, calculate costs, track active and completed orders, communicate with staff via messages, and update profile information. Staff and administrators can manage and update orders, verify payments, generate reports, manage user accounts, and upload project photos. The menu layout is designed for easy navigation, allowing users to quickly find and use the system's features.



3.3 Changing User ID and Password

To change your password, first log in to your account and click **Profile** from the menu. Then select **Change Password**, enter your current password, and type your new password twice to confirm. Finally, click **Save Changes** to update your password. Passwords must be at least eight characters long and include at least one uppercase letter, one lowercase letter, and one number. For security, avoid using easily guessable passwords such as “123456” or “password.”



3.4 Exit System

To log out, click Log Out either at the top right corner or at the bottom of the dashboard. You will be redirected to the homepage. Always remember to log out, especially when using public or shared computers, to keep your account secure.



4.0 USING THE SYSTEM

This section provides detailed descriptions and instructions for all major system functions. Menu items and features vary depending on whether the user is a Client, Staff, or Admin.

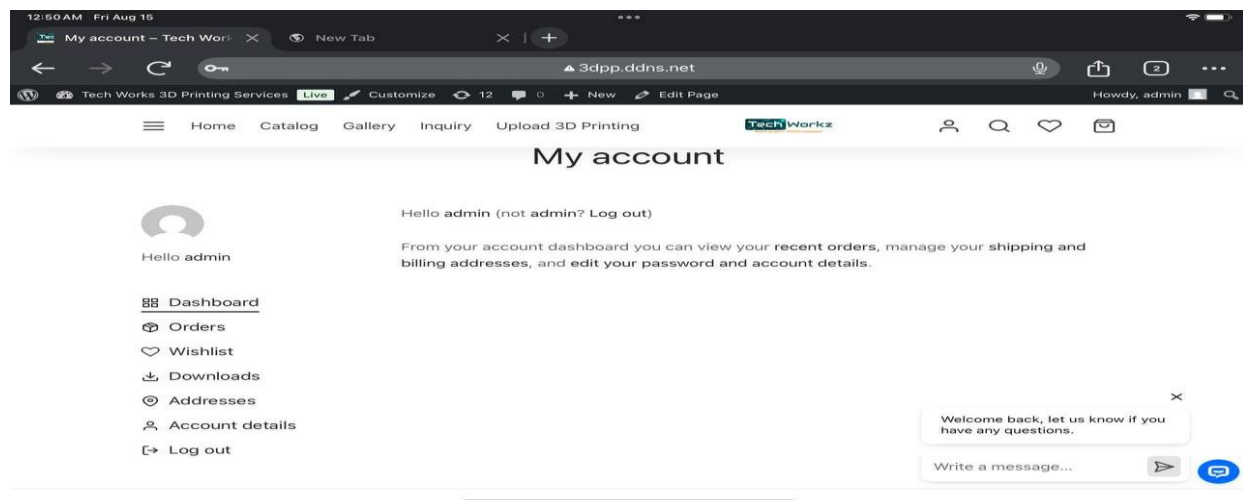
4.1 Dashboard

The Dashboard is the first page that appears after logging in and provides a quick overview of your account and recent activities. For clients, it displays active orders, pending payments, and recent messages, along with quick links to start a new order or view the gallery. For staff and administrators, it shows pending orders, recent payments, and alerts for tasks that require attention.

4.1.0 Client Dashboard

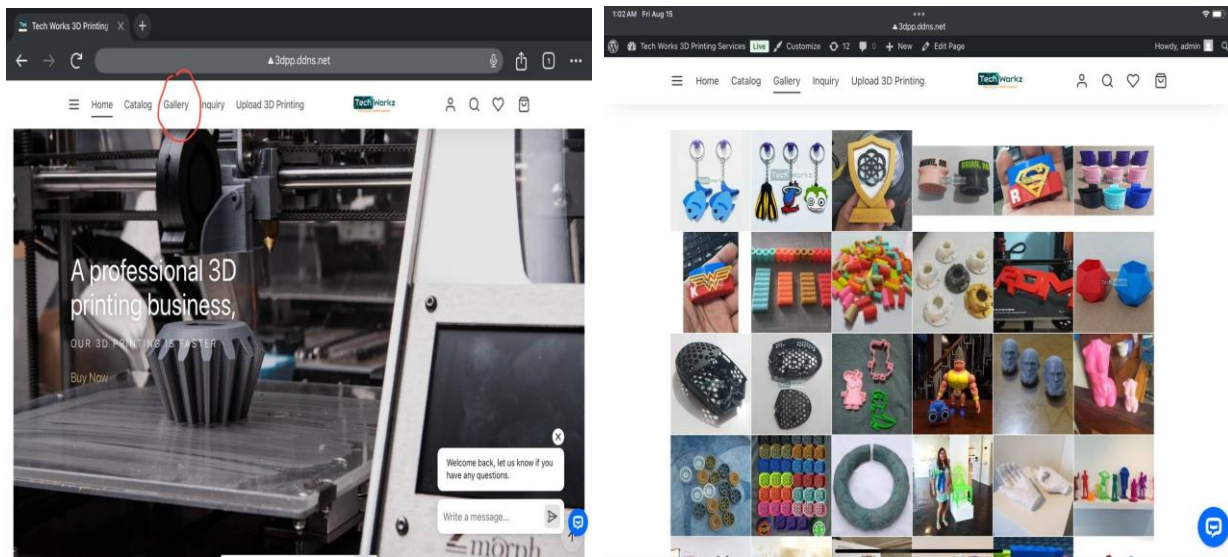


4.1.1 Staff and Administrators



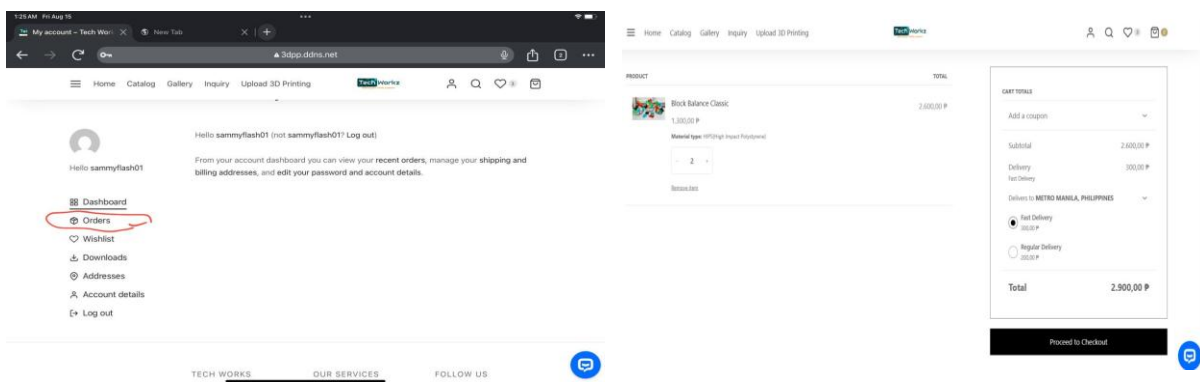
4.2 Gallery

The Gallery displays past projects and ready-to-print 3D designs. Users can access the Gallery from the system menu and browse photos or click on a project to view more details. Staff and administrators can upload new project images along with descriptions to keep the gallery updated.



4.3 Ready-to-Print Models

This section features pre-designed 3D models that clients can order instantly. To use this feature, clients can navigate to **Ready-to-Print Models** from the system menu, browse the available designs, and click **Order Now** to proceed with the order form.



4.0 Using The System

The first screenshot shows the 'Billing details' form with fields for First name (Sammy), Last name (Flash), Company name (optional), Country (Philippines), Street address (SM Shopping), Apartment/suite/unit/etc. (235), Town / City (optional), State / Country (Metro Manila), Postcode / ZIP (1118), Phone (09495214834), and Email address (sammyflash01@gmail.com). There is a checkbox for 'Ship to a different address?' and an 'Order notes (optional)' field.

The second screenshot shows the 'Place order' button and the 'SUBTOTAL' of 2,600.00 P. It also shows the 'SHIPPING' options: Fast Delivery (300.00 P) and Regular Delivery (300.00 P). The 'TOTAL' is 2,900.00 P. Payment methods include GCash via PayMongo, GCash, and Maya via PayMongo.

The third screenshot shows the 'Order details' for Order #4241, placed on August 14, 2025, and currently Processing. It lists the item 'Block Balance Classic - HIPS(High Impact Polystyrene)' with a material type of HIPS(High Impact Polystyrene) and a price of 2,600.00 P. The subtotal is 2,600.00 P, shipping is 300.00 P via Fast Delivery, and the payment method is GCash via PayMongo. The total is 2,900.00 P. It also shows the billing and shipping addresses for Sammy Flash.

4.4 Custom Order Submission

Clients can upload their own 3D model files for printing by selecting **Custom Order** from the menu. After uploading a supported file (STL or OBJ), they provide the required details such as dimensions, material type, and quantity, then click **Submit** to place the order.

The first screenshot shows the 'Upload 3D Printing' button in the top navigation menu, circled in red. The main content area features a large image of a 3D printer with the text 'A professional 3D printing business. OUR 3D PRINTING IS FASTER. Buy Now'.

The second screenshot shows the 'Upload 3D Printing' form. It includes fields for Name (Sammy Flash), Email (sammyflash01@gmail.com), and Phone (09397043434). There is a section for 'Upload 3D STL' with a 'Choose File' button and a 'File uploaded' status. A message field contains 'Hello Techworks'. A 'Send' button is circled in red at the bottom.

4.5 Cost Calculator

The Cost Calculator automatically estimates printing costs based on model details. Clients access it from the menu, enter the model's dimensions, material type, and quantity, and click **Calculate** to view the price.

Home Catalog Gallery Inquiry Upload 3D Printing

3D MODEL VIEWER
Upload 3D Model

3D Printing Cost Calculator

Cost of material

Material type: ABS (Acrylonitrile Butadiene Styrene)

Density: 1.05 g/cm³

Filament diameter: 1.75 mm

Filament length: mm

Filament weight: kg

Price: \$ / kg

Material cost: \$

Labor cost

Printing time: hrs

Cost per hour: \$ / hr

Labor cost: \$

Final price

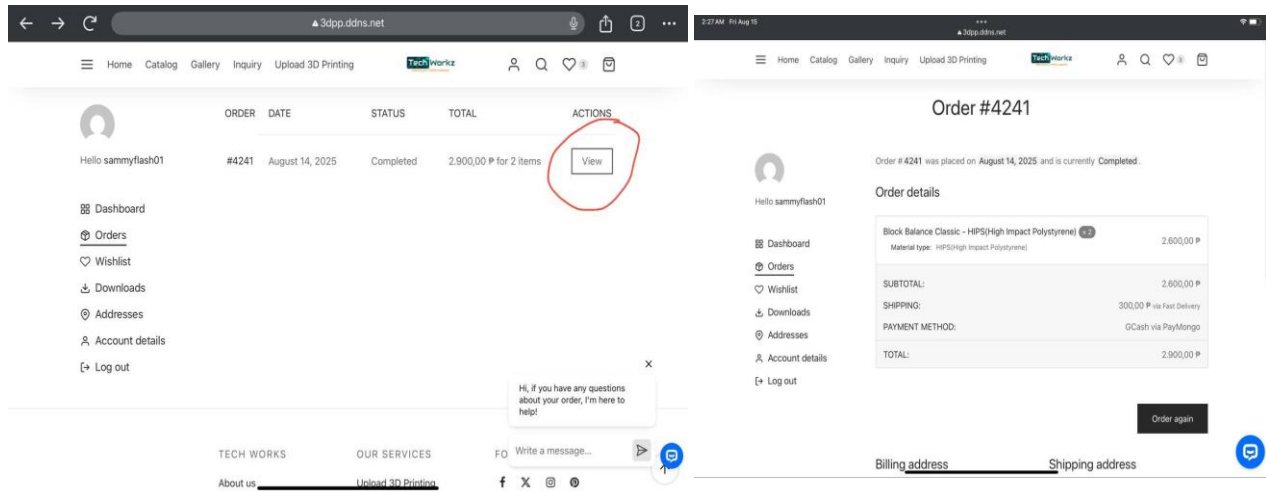
Markup: 10%

Final price: \$

Calculate

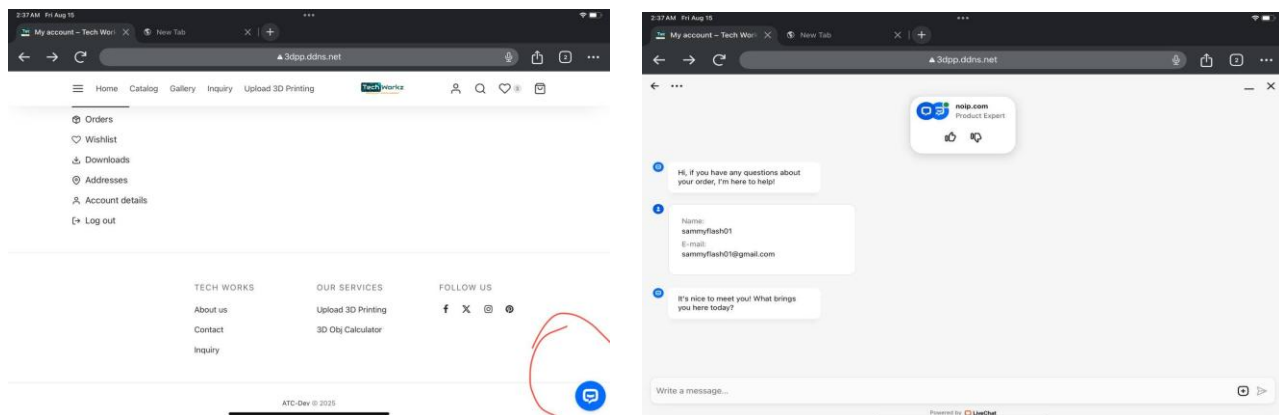
4.6 Order Tracking

Clients can monitor the progress of their orders in real-time through the **Order Tracking** section. Status updates include Pending Approval, Payment Verified, Printing in Progress, Ready for Pickup/Delivery, and Completed.



4.7 Messages/Chat

This feature allows direct communication between clients and TechWorkz staff. Clients access **Messages** from the menu, select an existing conversation or start a new one, type their message, and click **Send**. Staff and administrators can respond to messages and provide updates.



4.8 Payment & Verification

Clients make payments by selecting **Pay Now** on their order and choosing a payment method such as GCash, Maya, or Paymongo. They complete the transaction and upload proof if required. Staff and administrators verify payments by reviewing the uploaded proof and transaction reference, then clicking **Verify** to confirm the payment.

Edit Order

Activity

Finish setup

Edit order

Add new order

Order #4241 details

Payment via GCash via PayMongo (pay_yubGJssnHc2H6BXhYzndN). Paid on August 14, 2025 @ 5:23 pm.

Customer IP: 180.190.229.94

General

Billing

Shipping

Order attribution

Origin

Device type

Session page views

Customer history

Item	Cost	Qty	Total
Block Balance Classic - HIPS(High Impact Polystyrene)	1,300.00 ₱	x 2	2,600.00 ₱
SKU: block-balance-classic-version-7015563			
Variation ID: 3654			
Material type: HIPS(High Impact Polystyrene)			
Fast Delivery			300.00 ₱
Items: Block Balance Classic - HIPS(High Impact Polystyrene) x 2			
Items Subtotal:			2,600.00 ₱
Shipping:			300.00 ₱
Order Total:			2,900.00 ₱
Paid:			2,900.00 ₱

August 14, 2025 via GCash via PayMongo

4.9 Reports (Staff/Admin Only)

Staff and administrators can generate reports for sales, order history, and performance tracking. They select **Reports** from the menu, choose the report type (e.g., Daily Sales, Monthly Orders), set the date range, and click **Generate Report**.

Orders

Activity

Finish setup

Date range

Show

Month to date (Aug 1 - 15, 2025)

All orders

Orders

Net sales

Average order value

Average items per order

4

7,900.00 ₱

1,975.00 ₱

2

Orders

Month to date (Aug 1 - 15, 2025)

Previous year (Aug 1 - 15, 2024)

By day

Overview

Activity

Finish setup

Date range

Month to date (Aug 1 - 15, 2025)

vs. Previous year (Aug 1 - 15, 2024)

Performance

Total sales	Net sales	Orders	Products sold	Variations Sold
9,100.00 ₱	7,900.00 ₱	4	6	6

4.10 Special Instructions for Error Correction

For login issues, ensure a stable internet connection and correct credentials. File upload errors may occur if the file format or size is incorrect; only STL and OBJ formats are supported. Payment discrepancies should be reported to the TechWorkz Admin along with the transaction reference number.

4.11 Caveats and Exceptions

Orders will not start printing until payment has been verified. The system automatically logs out after 15 minutes of inactivity to ensure account security.

4.12 Input Procedures and Expected Output

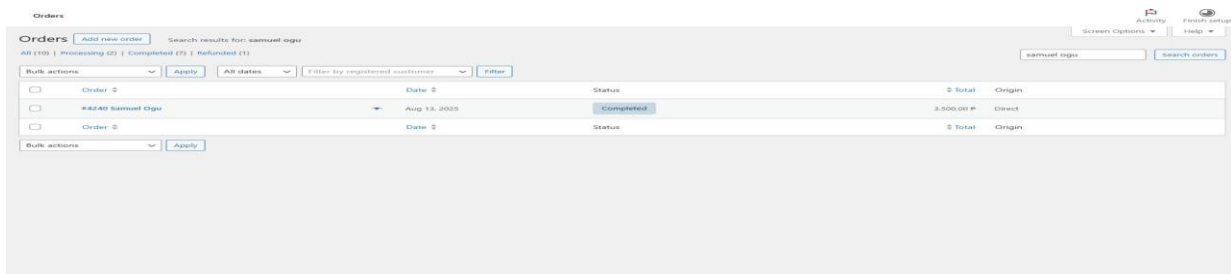
When placing an order, clients provide order details, upload files, or select a ready-made model as input. The system outputs an order confirmation page, sends an email notification, and updates the order status accordingly.

5.0 QUERYING

This section explains how staff and administrators can search for and retrieve specific information from the system. The query function allows quick access to orders, payments, and customer records.

5.1 Query Capabilities

The system supports searches for orders, clients, and payments. Orders can be located by Order ID, client name, date, or status, such as “Printing in Progress” or “Completed.” Clients can be searched by name, email, or registration date. Payment records can be retrieved using the payment reference number, client name, payment method, or payment status. Additionally, when generating reports, staff and administrators can filter data by date range, order status, or material type. All searches support partial matches, so typing part of a name or ID will still return results.



Order ID	Date	Status	\$ Total	Origin
#4240 Sammuel Ogu	Aug 13, 2025	Completed	3,500.00	Direct

5.2 Query Procedures

To search for an order, log in as staff or administrator, select **Orders** from the menu, enter the keyword in the search bar at the top right of the page, and press Enter or click the search icon. Matching results will appear in a table showing Order ID, client name, status, and date.

Dashboard

Posts

Media

Pages

Comments

Portfolio

Ninja Forms

GetWooFusion

WooCommerce

Home

Orders

Customers

Coupons

Reports

Settings

Status

Extensions

Products

Payments

Analytics

Marketing

Appearance

Orders

Add new order

Search results for: Block Balance Classic - HIPS(High Impact Polystyrene)

All (0) | Processing (2) | Completed (0) | Refunded (0)

Block Balance Classic - 3

Search orders

Bulk actions

Apply

All dates

Filter by registered customer

Filter

☐	Order #	Date	Status	Total	Origin
☐	#4241 Sammy Flash	2 hours ago	Completed	2,900.00 ₪	Direct
☐	#4955 John Aze Traavika	Jun 29, 2025	Refunded	2400.00 ₪ (1,00.00 ₪)	Direct
☐	Order #	Date	Status	Total	Origin

Bulk actions

Apply

6.0 REPORTING

The reporting feature allows staff and administrators to generate various reports for monitoring business performance, tracking orders, and managing finances. Reports can be filtered by date range, order status, and material type to focus on relevant information.

6.1 Types of Reports Available

The system provides several types of reports. The **Daily Sales Report** lists all completed orders for the day, including order IDs, client names, payment amounts, and payment methods. The **Monthly Sales Report** summarizes sales for the month, showing total revenue, completed orders, and best-selling materials. The **Order History Report** displays all orders within a selected time frame, including both completed and pending transactions. The **Customer Activity Report** shows customer engagement, including the number of orders, purchase frequency, and order value. Optionally, the **Material Usage Report** tracks the quantities of materials such as PLA and ABS used during the selected period.

6.2 Generating a Report

To generate a report, log in as staff or administrator and select **Reports** from the menu. Choose the report type, set the start and end dates, apply additional filters if needed (such as order status or material type), and click **Generate Report**.



Curriculum Vitae

Name: Abdulaziz Abdelnour Mohammed Fitwi

Address: Sta. Ana Manila

Contact Numbers:09559346435



CAREER OBJECTIVE

What are your goals?

"I want to work my way through an organization and become a leader with clear goals and someone who can contribute to the organization's success

PERSONAL INFORMATION

Date of Birth: July/13/1996

Sex: Male

Civil Status: Single

Citizenship. Eritrean

Descriptions you want to highlight

Proven ability to work collaboratively with diverse groups and establish ongoing relationships.
Capable of quickly understanding how systems are used in order to provide efficient support.
Demonstrated outstanding customer orientation using strong communication skills.

EDUCATIONAL ATTAINMENT

Bachelor's Degree: BS INFORMATION TECHNOLOGY

- University , AMA UNIVERSITY
- 4Year Undergraduate
- Address : Quezon City Panay Ave

Thesis Title

- Secondary: Al'iibrahimia school
- School, Year Graduated 2013
- Address Saudi Arabia Jeddah city Hail ST

SEMINARS / TRAININGS ATTENDED

WebDesign / Office application
GMS TECHNOLOGY 200
Detailed Description of the Seminar / Training
learning knowledge and skills through direct practice.

REFERENCES

Name

Position: Call center agent , Company: concentric
Email address: mf.soriano007@gmail.com
Contact numbers: 0915 476 9257

Samuel Chisom Ogu

Information Technology Graduate

Entry-Level Cybersecurity & IT Infrastructure Professional

samuelogu01@gmail.com | +639495214834

<https://www.linkedin.com/in/samuel-ogu/>

Quezon City, Metro Manila, 1118, Philippines



PROFESSIONAL SUMMARY

Motivated and detail-oriented Information Technology graduate with a strong foundation in Information Security and IT Infrastructure. Gained practical experience through coursework, internship, and industry-recognized certifications from Google, IBM, Microsoft, Cisco, and EC-Council. Developed a secure web-based system for the capstone project, applying secure coding and deployment practices. Skilled in network support, system administration, and basic cybersecurity operations. Eager to apply technical knowledge to solve real-world IT and security challenges.

EDUCATION

AMA University

Bachelor of Science in Information Technology

Quezon City, Philippines

Expected Completion: October 2025

Relevant Coursework: Information Assurance and Security, Data Communications and Networking, System Administration and Maintenance, Cloud Computing, Software Engineering, System Integration and Architecture, Database Management Systems, Operating Systems, Application Development, Project Management, Capstone Project, IT Practicum

EXPERIENCE

First Oceanic Property Management Inc. (FOPM),

Information Security & IT Infrastructure Intern

Taguig, Philippines

May 2023 - August 2023

- Resolved 95% of diverse help desk issues using ServiceNow and remote support tools, significantly improving IT support performance and response times.
- Reduced internal security incidents by 40% through strategic firewall configuration and continuous network monitoring.
- Assisted in endpoint security updates and enforced access control protocols to align with INFOSEC best practices.
- Conducted threat analysis using Wireshark and monitored system health using PRTG, contributing to improved network visibility.
- Participated in backup and recovery implementation, ensuring business continuity and data integrity.

AIG Shields Up: Cybersecurity virtual experience program on Forage

January 2025

- Completed threat simulation based on CISA guidance
- Conducted vulnerability analysis and scripted decryption with ethical hacking using Python

PROJECT

AMA University Capstone Project: 3D Printing Service Ordering Management System November 2024

- Developed a secure web application using PHP, WordPress, HTML/CSS, and JavaScript.
- Integrated user authentication, data validation, and session handling to reduce risk of XSS and SQL injection.
- Deployed system using SmarterASP.net cloud hosting and designed database with MySQL.
- Improved service efficiency and streamlined secure online order tracking.

CERTIFICATIONS

Core Cybersecurity Certificate:

- Google Cybersecurity Specialization - Google (Dec 2023)
- Microsoft Cybersecurity Analyst Specialization - Microsoft (Jan 2024)

Samuel Chisom Ogu

Information Technology Graduate

Entry-Level Cybersecurity & IT Infrastructure Professional

samuelogu01@gmail.com | +639495214834

<https://www.linkedin.com/in/samuel-ogu/>

Quezon City, Metro Manila, 1118, Philippines



- Cisco Cybersecurity Operations Fundamentals - Cisco Networking Academy (Apr 2024)
- Cybersecurity Attack & Defense - EC-Council (May 2024)
- IBM Cybersecurity Analyst Assessment - IBM (Aug 2024)
- AIG Shields Up Cybersecurity Simulation - Forage (Jan 2025)

Cloud & Platform Security:

- Introduction to Information Technology and AWS Cloud (Feb 2024)
- IBM Generative AI for Cybersecurity - IBM (Jun 2024)
- Security in Google Cloud - Google Cloud (Jul 2024)
- Palo Alto Networks Cybersecurity - Palo Alto (Aug 2024)
- AI for Cybersecurity - Johns Hopkins University (Jun 2025)

Data & IT Foundations:

- IBM IT Support Specialization - IBM (Jan 2024)
- Google Data Analytics - Google (May 2024)
- Cybersecurity in Healthcare - Erasmus University Rotterdam (May 2024)

TECHNICAL SKILLS

Cybersecurity & GRC:

- Threat detection & incident response, basic SIEM knowledge, Wireshark, Nessus, firewall & access control, risk identification, security policy documentation (CISA, ISO 27001 aligned), Malware Analysis, Threat Intelligence, Threat Hunting, Log Analysis

IT Infrastructure:

- Windows, MacOS, Linux troubleshooting, Active Directory, help desk support (ServiceNow), system backup & recovery, endpoint security, remote desktop tools

Cloud & Virtualization:

- Google Cloud (IAM, VPC, cloud monitoring), AWS & Azure basics, VMware, Hyper-V, smarterasp.net deployment

Programming & Tools:

- Python (scripting, automation), SQL, Java, PHP, HTML/CSS, Cisco Packet Tracer, Microsoft Office, Google Workspace, Programming and Scripting, Network Security, Incident Response, Compliance and Risk Management

SOFT SKILLS

- Collaboration, accountability, effective communication in high-pressure environments, Systems Design and Strategic Thinking, Creativity, Continuous Learning

RESEARCH INTEREST

- AI-driven Security Automation for Health Information Systems
- Secure Network Architecture for Healthcare Data Protection
- Cybersecurity in Organizational IT Transformation
- Risk-based Vulnerability Management for Public Health Infrastructure
- Innovation in End-user Security Awareness and Governance

LANGUAGES:

- **English** - Fluent (Academic and Professional Proficiency)
- **Igbo** - Native
- **Tagalog** - Basic Conversational
- **Spanish** - Basic Conversational